22050 Signals and linear systems in continuous time

Kaj-Åge Henneberg

L02

Time-domain Impulse response

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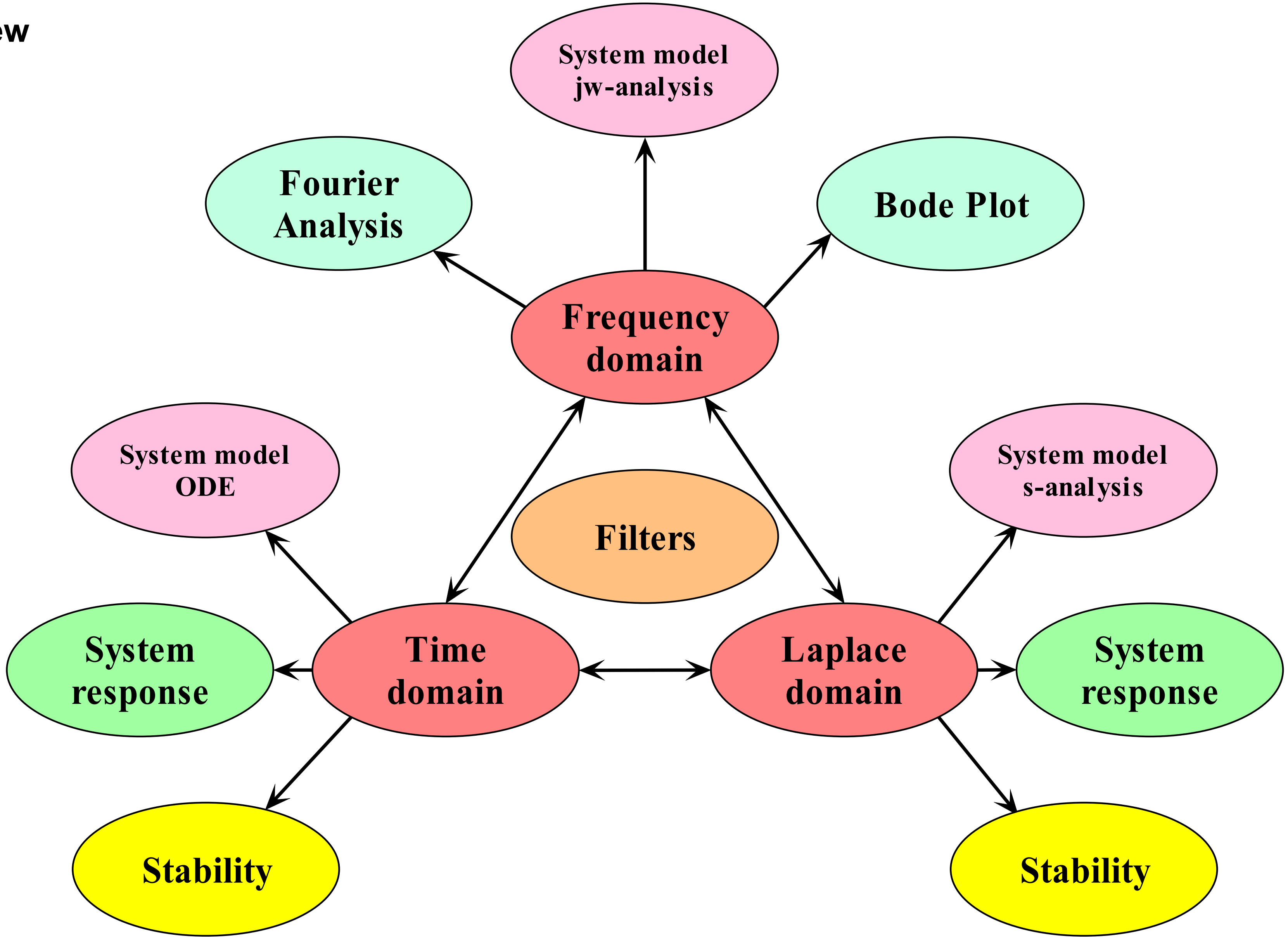
Course overview

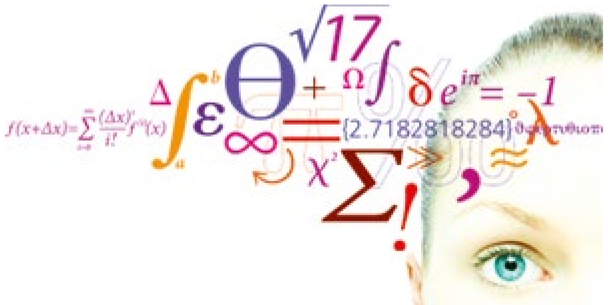
- System
  - Model
  - Response
  - Performance

- Domains
  - Time
  - Frequency
  - s-domain



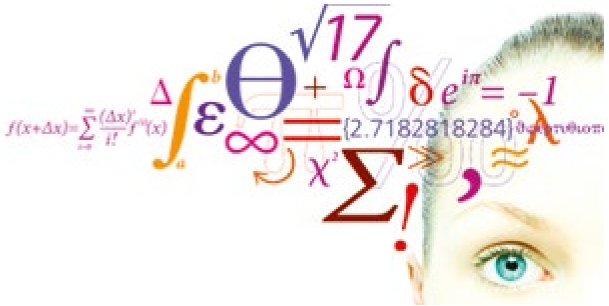
Lecture Plan

- System equation for mechanical system
- Decomposition property
- Zero-input response
- Unit impulse response
- Examples



Lathi: 2.1 – 2.3 + 2.8

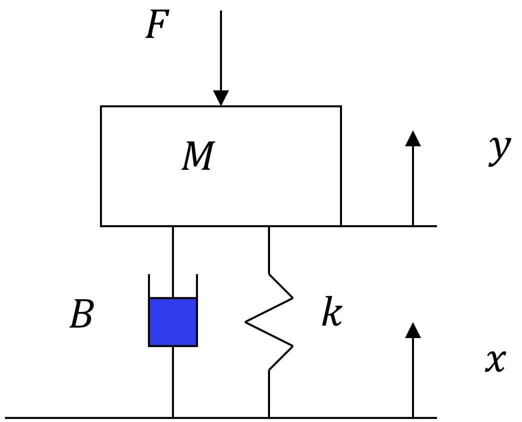
Next topic



Decomposition property

Video 1

Wheel suspension model



Mass M could be a car, k its springs, and B its shock absorbers.

Ground level is $x(t)$ and the level of a marked point on the car is $y(t)$.

ΔL: Spring compression due to gravity

$$\underbrace{M\ddot{y}}_{Acceleration} = -\underbrace{F}_{Force} - \underbrace{Mg}_{Gravity} - \underbrace{k(y - x - \Delta L)}_{Spring} - \underbrace{B(\dot{y} - \dot{x})}_{Damper}$$

$$M\ddot{y} + B\dot{y} + ky = -F - Mg + B\dot{x} + kx + k\Delta L$$

At rest: $y = \dot{y} = \ddot{y} = x = \dot{x} = F = 0 \Rightarrow Mg = k\Delta L$

$$M\ddot{y} + B\dot{y} + ky = -F + B\dot{x} + kx$$

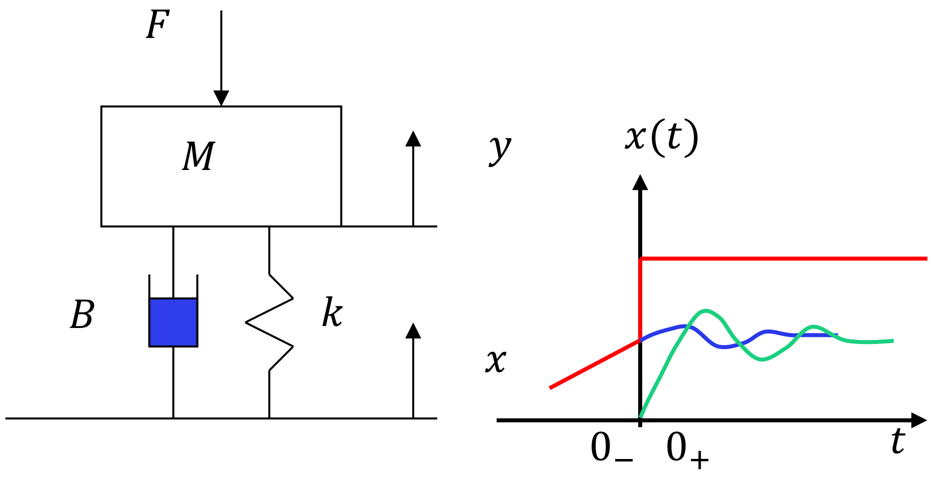
Zero force: $\ddot{y} + \frac{B}{M}\dot{y} + \frac{k}{M}y = \frac{B}{M}\dot{x} + \frac{k}{M}x$ Standard form

Initial conditions

There are different ways to define the initial conditions. We need to understand the difference and their pros and cons.

Let us think of the suspension of a car passing a sudden increase in elevation. We will assume this happens at $t = 0$.

We want to describe the elevation $y(t)$ of the car after passing the sudden elevation.



$$\ddot{y} + \frac{B}{M}\dot{y} + \frac{k}{M}y = \frac{B}{M}\dot{x} + \frac{k}{M}x$$

Classical solution of differential equations: $y(0_+) = y_0, \qquad \dot{y}(0_+) = \dot{y}_0$

The shortcoming of this method is that we don't know how much the jump in ground elevation (input $x(t)$) contributed to this new state and how much the state of the car suspension before the jump $(y(0_-), \dot{y}(0_-))$ contributes to the new state. We cannot separate out the energy in the new state into what energy was in the system before the jump and how much came from the jump.

New approach: $y(0_-) = y_0, \qquad \dot{y}(0_-) = \dot{y}_0$

In this approach, we specify the state of the system, before the jump. This state is the result of all previous $(t < 0)$ changes in ground level elevation but is completely independent of what happens for $t \geq 0$. The state (energy) of the system at $t = 0_-$ will dissipate and settle at its equilibrium elevation (blue curve). Independent of this, the car will react to the jump in elevation (green curve). The advantage of this approach is that we can develop methods to compute the response to any arbitrary changes in input measured by sensors, not just inputs stated as mathematical equations.

Decomposition property

This mechanical system will change

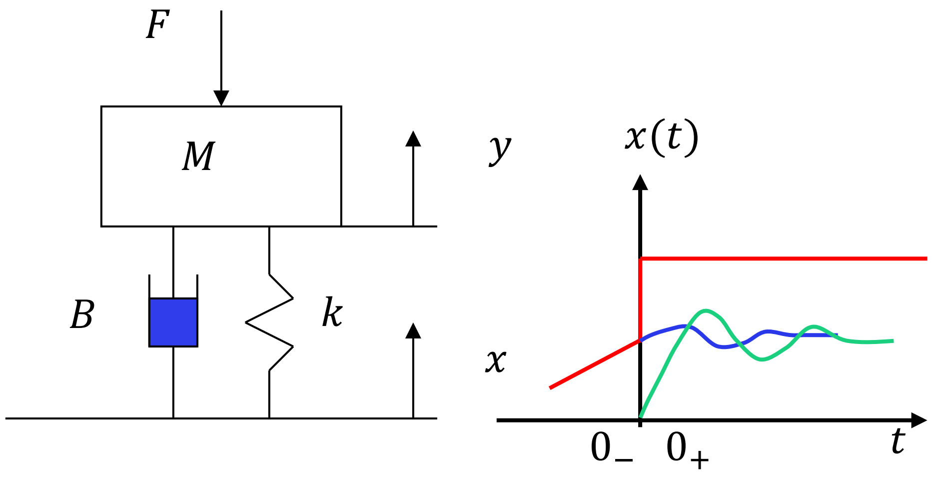
its state at $t > 0$ due to three cause:

Zero-input response

1. The state is allowed to dissipate to its resting equilibrium at $t > 0$ after an initial displacement from rest. This produces a transient response. Blue curve.

Zero-state response

2. The ground level x is displaced. This produces an **input-driven response**. Green curve.



3. A combination of 1 and 2.

Total response = zero-input response (internal energy)

+ zero-state response (external energy)

Decomposition is based on the principle of superposition; thus, **decomposition is only possible for linear systems**.

Operator format

Standard form:
n'th order

$$\frac{d^n y}{dt^n} + a_{n-1}\frac{d^{n-1}y}{dt^{n-1}} + \ldots + a_1\frac{dy}{dt} + a_0 y(t) = b_m\frac{d^m x}{dt^m} + b_{m-1}\frac{d^{m-1}x}{dt^{m-1}} + \ldots + b_1\frac{dx}{dt} + b_0 x(t)$$

Operator form:

$$(D^n + a_{n-1}D^{n-1} + \ldots + a_1 D + a_0)y(t) = (b_m D^m + b_{m-1}D^{m-1} + \ldots + b_1 D + b_0)x(t)$$

Generic form:

$$Q(D)y(t) = P(D)x(t)$$

To avoid amplifying noise in the input signal, the order of the polynomial P should not be higher than Q, i.e. $m \leq n$.

Example:

$$Q(D) = D^2 + \frac{1}{\tau_1 + \tau_2}D + \frac{k}{\tau_1\tau_2} \qquad P(D) = \frac{1}{\tau_1 + \tau_2}D + \frac{k}{\tau_1\tau_2}$$

$$n = 2 \qquad m = 1$$

Decomposition property

Decomposition is based on the principle of superposition.

Thus, decomposition is only possible for linear systems, i.e., where $Q(D)$ and $P(D)$ are linear operator polynomials.

In terms of techniques, we still need to be able to solve differential equations, and the methods are almost identical. But the new approach also allows us to apply new methods.

Total response = zero-input response (internal energy)

+ zero-state response (external energy)

$$y(t) = \underbrace{y_{zi}(t)}_{\substack{zero\\input}} + \underbrace{y_{zs}(t)}_{\substack{zero\\state}}$$

$$Q(D)y(t) = P(D)x(t)$$

$$Q(D)\big(y_{zi}(t) + y_{zs}(t)\big) = P(D)x(t)$$

$$Q(D)y_{zi}(t) + Q(D)y_{zs}(t) = P(D)x(t)$$

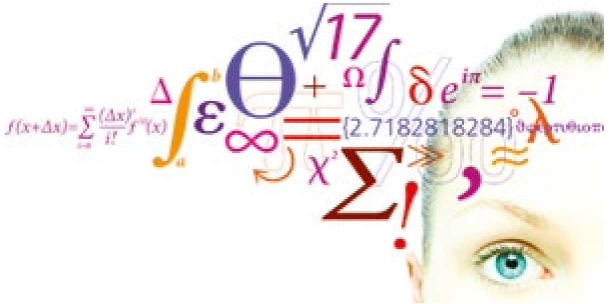
$$Q(D)y_{zi}(t) = 0$$

Homogeneous differential equation

$$Q(D)y_{zs}(t) = P(D)x(t)$$

Inhomogeneous differential equation with zero initial conditions

Next topic



Zero-input response

Video 2

Zero-input response

In order that the left-hand side can equal zero for all times t, the solution to the homogeneous equation, $y_0(t)$, and all its derivatives must have the same form, that is:

$$(1): (D^n + a_{n-1}D^{n-1} + \ldots + a_1 D + a_0)y(t) = 0$$

$$y_0(t) = Ce^{\lambda t}$$

Inserting in (1) yields:

$$C(\lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0)e^{\lambda t} = 0$$

Because $e^{\lambda t}$ is outside the parenthesis, the equation can equal zero for all times only if the content inside the parenthesis is zero.

Characteristic equation and its roots:

$$Q(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \ldots + a_1\lambda + a_0 = 0$$
$$= (\lambda - \lambda_n)(\lambda - \lambda_{n-1}) \ldots\ldots (\lambda - \lambda_1) = 0$$

Natural modes: $C_1 e^{\lambda_1 t}, C_2 e^{\lambda_2 t} \ldots\ldots, C_n e^{\lambda_n t}$

Natural response types

1. If n different real roots:

$$y_0(t) = \sum_{j=1}^{n} C_j e^{\lambda_j t}, t \geq 0$$

$\lambda_j < 0$, exponentially decreasing response

$\lambda_j > 0$, exponentially increasing response

2. If r repeated real roots:

$$y_0(t) = (C_1 + C_2 t + \ldots + C_r t^{r-1}) e^{\lambda_1 t} + C_{r+1} e^{\lambda_{r+1} t} + \cdots + C_n e^{\lambda_n t}$$

3. If complex conjugated roots:

For a polynomial $Q(\lambda)$ with real coefficients, complex roots will always appear in conjugated pairs.

$$\begin{aligned} y_0(t) &= C_1 e^{(\sigma + j\omega)t} + C_2 e^{(\sigma - j\omega)t} \\ &= C_a e^{\sigma t} \cos(\omega t) + C_b e^{\sigma t} \sin(\omega t) \\ &= C_c e^{\sigma t} \cos(\omega t + \theta) \end{aligned}$$

Three typical forms

$$\ddot{y}(t) + a_1 \dot{y}(t) + a_0 y(t) = 0;$$

$$\ddot{y}(t) + 2\sigma \dot{y}(t) + \sigma^2 y(t) + \omega^2 y(t) = 0;$$

$$\sigma = \frac{a_1}{2} \qquad \omega^2 = a_0 - \sigma^2$$

Addition of sinusoidal functions

$$C\cos(\omega\, t+\theta) = C\cos(\theta)\cos(\omega\, t) - C\sin(\theta)\sin(\omega\, t) = a\cos(\omega\, t) + b\sin(\omega\, t)$$

$$a = C\cos(\theta)\,, b = -C\sin(\theta)$$

$$C = \sqrt{a^2+b^2}$$

$$\theta = \tan^{-1}\left(-\frac{b}{a}\right)$$

Determination of coefficients

Example:

System with two real distinct roots

We are here only concerned with homogeneous diff. equations. Hence there are no particular solution here.

n unknown coefficients require *n* initial conditions

$$y_0(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$
$$\dot{y}_0(t) = C_1 \lambda_1 e^{\lambda_1 t} + C_2 \lambda_2 e^{\lambda_2 t}$$

$$C_1 + C_2 = y_0(0_-)$$

$$C_1 \lambda_1 + C_2 \lambda_2 = \dot{y}_0(0_-)$$

$$C_2 = \frac{\lambda_1 y_0(0_-) - \dot{y}_0(0_-)}{\lambda_1 - \lambda_2}$$

$$C_1 = y_0(0_-) - C_2 = -\frac{\lambda_2 y_0(0_-) - \dot{y}_0(0_-)}{\lambda_1 - \lambda_2}$$

Example 1: Two real roots

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 0; y(0_-) = 2, \dot{y}(0_-) = 1$$

$$Q(\lambda) = \lambda^2 + 3\lambda + 2 = 0 \qquad \frac{3^2}{2} > 4 \Leftrightarrow real\ roots$$

$$Q(\lambda) = (\lambda + 2)(\lambda + 1) = 0$$
$$\Leftrightarrow \lambda_1 = -2 \wedge \lambda_2 = -1$$

$$y_0(t) = C_1 e^{-2t} + C_2 e^{-t}; t \geq 0$$

$$C_2 = \frac{\lambda_1 y_0(0_-) - \dot{y}_0(0_-)}{\lambda_1 - \lambda_2} = \frac{-2 \cdot 2 - 1}{-2 - (-1)} = 5$$

$$C_1 = y_0(0_-) - C_2 = 2 - 5 = -3$$

$$y_0(t) = 5e^{-t} - 3e^{-2t}; t \geq 0$$

Initial conditions

$ics := y(0) = 2, \mathrm{D}(y)(0) = 1$

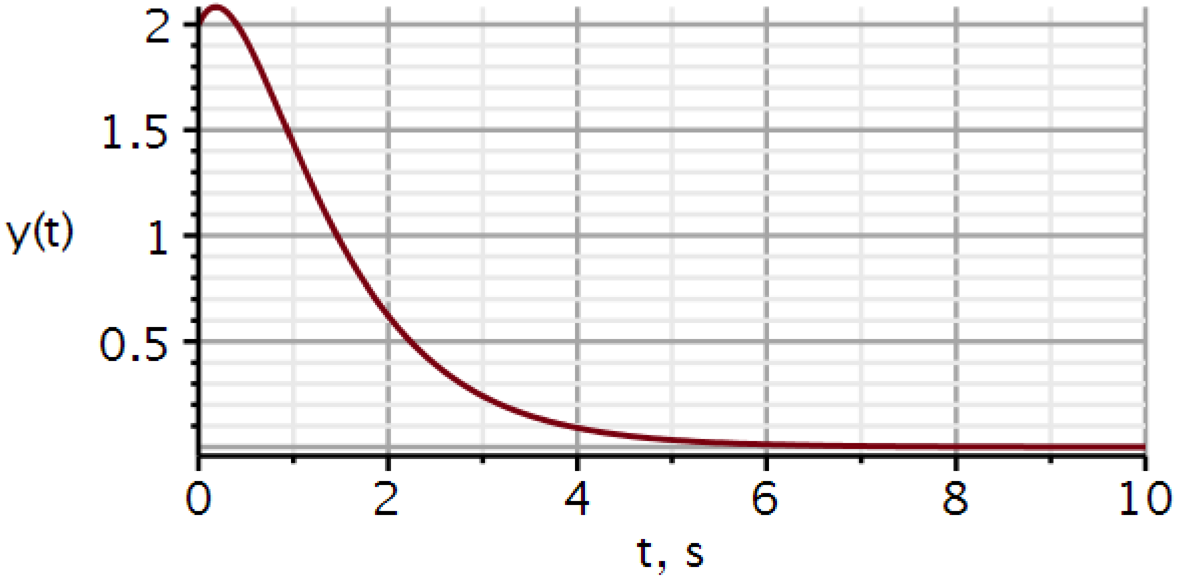
$ics := y(0) = 2, \mathrm{D}(y)(0) = 1$

Differential equation

$ode := diff(y(t), t, t) + 3 \cdot diff(y(t), t) + 2 \cdot y(t) = 0 :$

$solution := evalf(dsolve(\{ode, ics\}, y(t)))$

$solution := y(t) = 5.\ \mathrm{e}^{-1.\ t} - 3.\ \mathrm{e}^{-2.\ t}$



Example 2: A double root

$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = 0; y(0_-) = 2, \dot{y}(0_-) = 1$$

$$Q(\lambda) = \lambda^2 + 6\lambda + 9 = 0 \qquad \frac{6^2}{9} = 4 \Leftrightarrow double\ roots$$

$$Q(\lambda) = (\lambda + 3)(\lambda + 3) = 0 \Leftrightarrow \lambda = -3$$

$$y_0(t) = (C_1 + C_2 t)e^{-3t}; t \geq 0$$

$$y_0(0_-) = C_1 = 2$$

$$\dot{y}_0(t) = (C_1\lambda + C_2 + C_2\lambda t)e^{\lambda t}$$

$$\dot{y}_0(0_-) = C_1\lambda + C_2 \Rightarrow C_2 = \dot{y}_0(0_-) - C_1\lambda$$
$$C_2 = 1 - 2(-3) = 7$$

$$y_0(t) = (2 + 7t)e^{-3t}; t \geq 0$$

Initial conditions

ics := $y(0) = 2$, $D(y)(0) = 1$

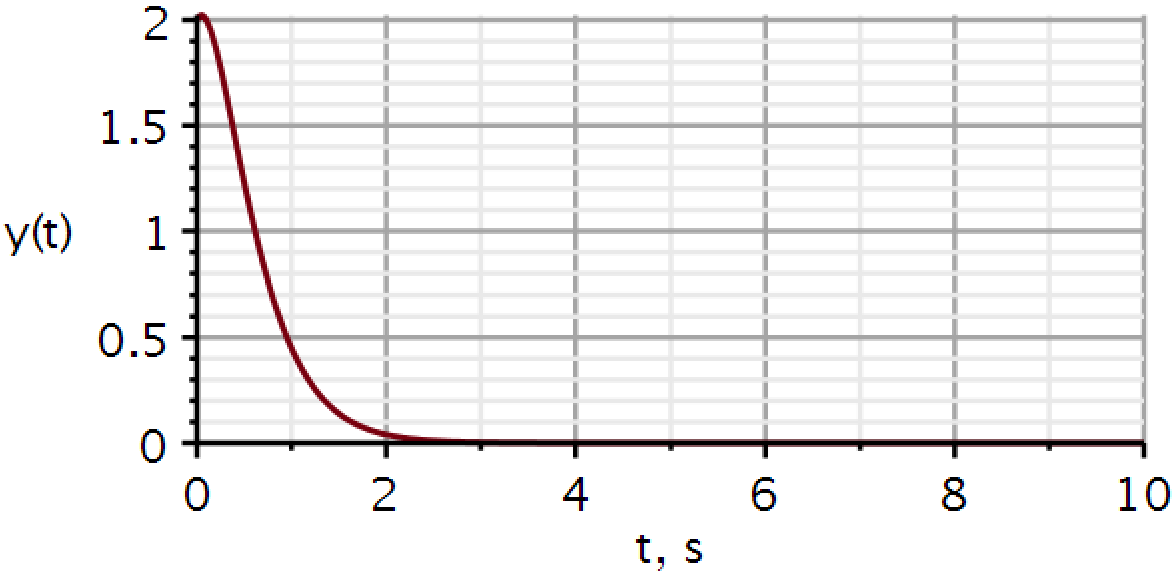
ics := $y(0) = 2$, $D(y)(0) = 1$

Differential equation

ode := *diff*$(y(t), t, t) + 6 \cdot$ *diff*$(y(t), t) + 9 \cdot y(t) = 0$:

solution := *evalf*(*dsolve*({*ode*, *ics*}, $y(t)$))

solution := $y(t) = 2.\, e^{-3.\, t} + 7.\, e^{-3.\, t}\, t$



Example 3: Complex conjugated roots

$$\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = 0; y(0\_) = 2, \dot{y}(0\_) = 1$$

$$Q(\lambda) = \lambda^2 + 4\lambda + 13 = 0 \qquad \frac{4^2}{13} < 4 \Leftrightarrow comp\ conj\ roots$$

$$Q(\lambda) = (\lambda + 2 + j3)(\lambda + 2 - j3) = 0 \Leftrightarrow \lambda = -2 \pm j3$$

$$y_0(t) = C_1 e^{-2t}\cos(3t) + C_2 e^{-2t}\sin(3t)\,; t \geq 0$$

$$y_0(0\_) = C_1 = 2 \qquad \sigma = \frac{a_1}{2} = 2 \qquad \omega^2 = a_0 - \sigma^2 = 9$$

$$\dot{y}_0(t) = C_1\sigma e^{\sigma t}\cos(\omega t) - C_1\omega\sin(\omega t)$$
$$+C_2\sigma e^{\sigma t}\sin(\omega t) + C_2\omega e^{\sigma t}\cos(\omega t)$$

$$\dot{y}_0(0\_) = C_1\sigma + C_2\omega = 1 \Rightarrow C_2 = \frac{\dot{y}_0(0\_) - C_1\sigma}{\omega}$$

$$\Rightarrow C_2 = \frac{1 - 2(-2)}{3} = \frac{5}{3}$$

$$y_0(t) = 2e^{-2t}\cos(3t) + \frac{5}{3}e^{-2t}\sin(3t)\,; t \geq 0$$

Initial conditions

$ics := y(0) = 2, \mathrm{D}(y)(0) = 1$

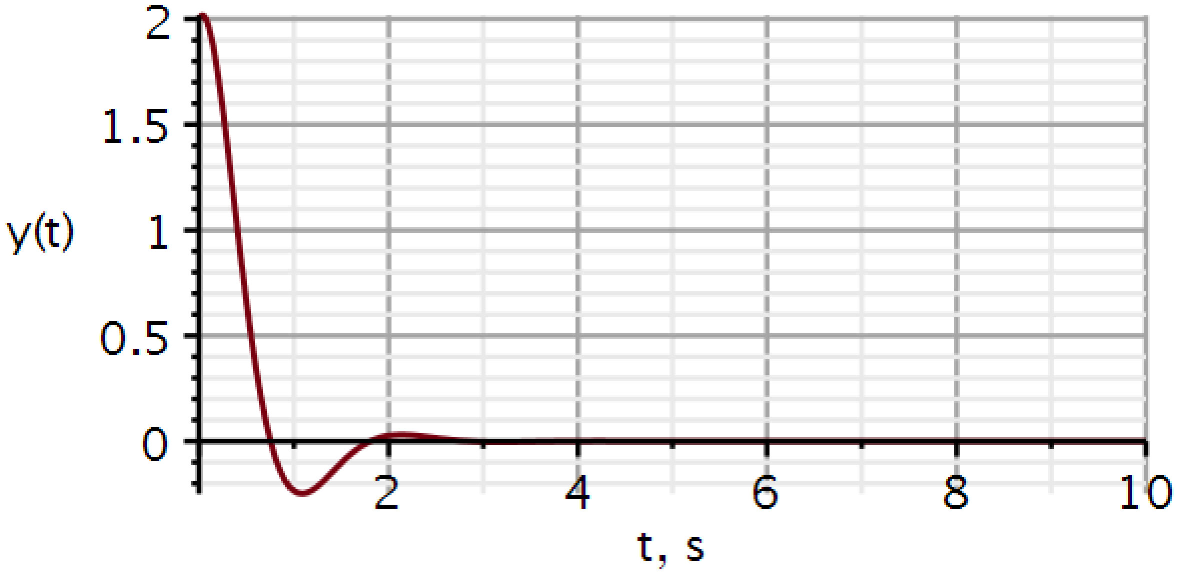
$ics := y(0) = 2, \mathrm{D}(y)(0) = 1$

Differential equation

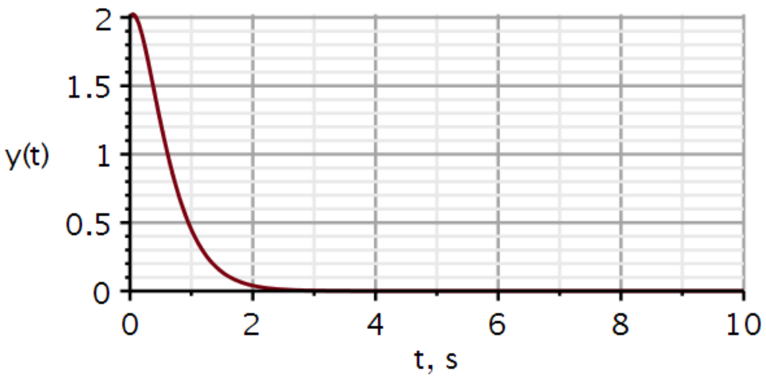
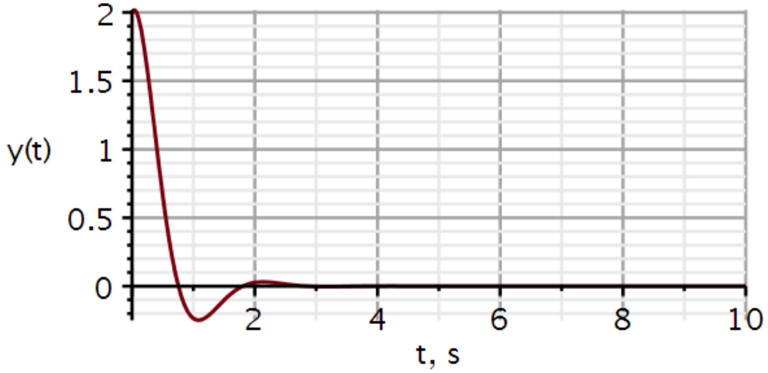
$ode := diff(y(t), t, t) + 4 \cdot diff(y(t), t) + 13 \cdot y(t) = 0 :$

$solution := evalf(dsolve(\{ode, ics\}, y(t)))$

$solution := y(t) = 1.666666667\, \mathrm{e}^{-2.\, t} \sin(3.\, t) + 2.\, \mathrm{e}^{-2.\, t} \cos(3.\, t)$



Variation in dynamics

Two real roots	A double root	Complex conjugated roots
		

Slow return to resting state

Largest time constant slows down decay

Fast return to resting state

Fastest dissipation but overshoots and oscillates

The roots of the characteristic equation define the nature of the system.

Maple package: DynamicSystems

We will use Maple throughout the course and use some relevant tools. One package of high relevance is **DynamicSystems**:

$$u: input\ signal$$
$$x: system\ state$$
$$y: output\ signal$$
$$t: time$$
$$s: complex\ frequency: \sigma + j\omega$$

Normally we use $u(t)$ to denote the step function. When using DynamicSystems, $u(t)$ is reserved to denote input. In this situation I use $v(t)$ to denote the step function in Maple scripts.

```
> with(DynamicSystems) :
Return a sequence of all options and their values.
> SystemOptions( )
inputvariable = u, statevariable = x, discretefreqvar = z, relativeerror
   = 0.001, colors = ["#78000E", "#000E78", "#4A7800", "#3E578A",
  "#780072", "#00786A", "#604191", "#004A78", "#784C00", "#91414A",
  "#3E738A", "#78003B", "#00783F", "#914186", "#510078", "#777800"]
  , duration = 10.0, radians = false, discrete = false, linearsolvemethod
   = none, cancellation = false, conjugate = false, decibels = true,
  discretetimevar = q, samplecount = 10, outputvariable = y,
  continuoustimevar = t, parameters = ∅, complexfreqvar = s, hertz = false,
  sampletime = 1.
```

Maple: Defining systems

There are several ways to define a system.

In the example shown here, a function named **Coefficients** are used. It takes as input:

$$\frac{P(s)}{Q(s)}$$

In Maple, "*s*" denotes the differential operator.

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = x(t)$$

$$s^2 y(t) + 3sy(t) + 2y(t) = x(t) \Rightarrow y(t) = \frac{1}{s^2 + 3s + 2} x(t)$$

$$\frac{P(s)}{Q(s)} = \frac{1}{s^2 + 3s + 2}$$

Dynamic systems

$restart$

$with(plots):$

$with(DynamicSystems):$

$\boxed{sys1} := Coefficients\left(\frac{1}{s^2 + 3 \cdot s + 2}\right):$

$PrintSystem(sys1)$

$$\left[\begin{array}{l} \textbf{Coefficients} \\ \text{continuous} \\ \text{1 output(s); 1 input(s)} \\ \text{inputvariable} = [u1(s)] \\ \text{outputvariable} = [y1(s)] \\ \text{num}_{1,1} = [1] \\ \text{den}_{1,1} = [1, 3, 2] \end{array}\right.$$

Maple: Plotting roots of numerator and denominator operators

$$\boxed{ZeroPolePlot}(sys1, symbolsize = 30, font = [Helvetica, roman, 18], axis[2] = [thickness = 2.0], axis[1] = [mode = linear, thickness = 2.0], size = [400, 400])$$

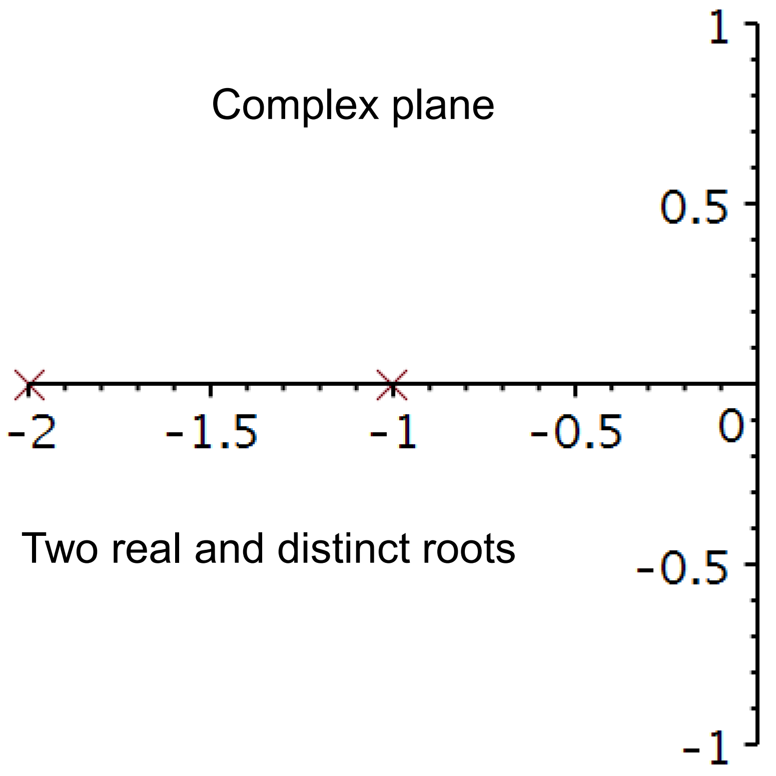
$$Q(D)y(t) = P(D)x(t)$$

$$y(t) = \frac{P(D)}{Q(D)}x(t)$$

The roots of $P(D) = 0$ are called "**ZEROS**"

The roots of $Q(D) = 0$ are called "**POLES**"

The function **ZeroPolePlot** will plot the zeros and poles for a predefined system, here named *sys1*.



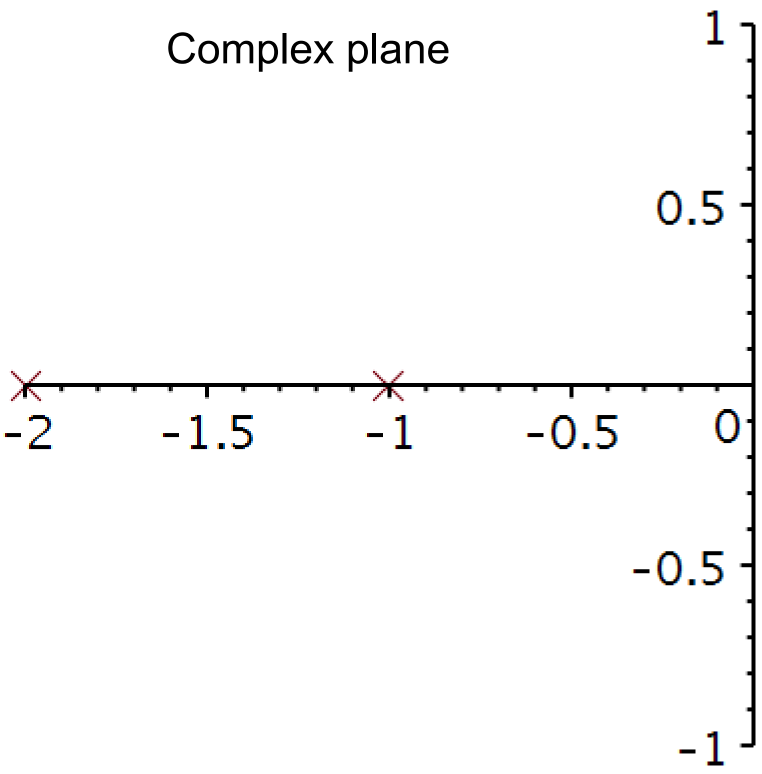
Observation

The poles are also the roots of the characteristic equation.

Roots of characteristic equations

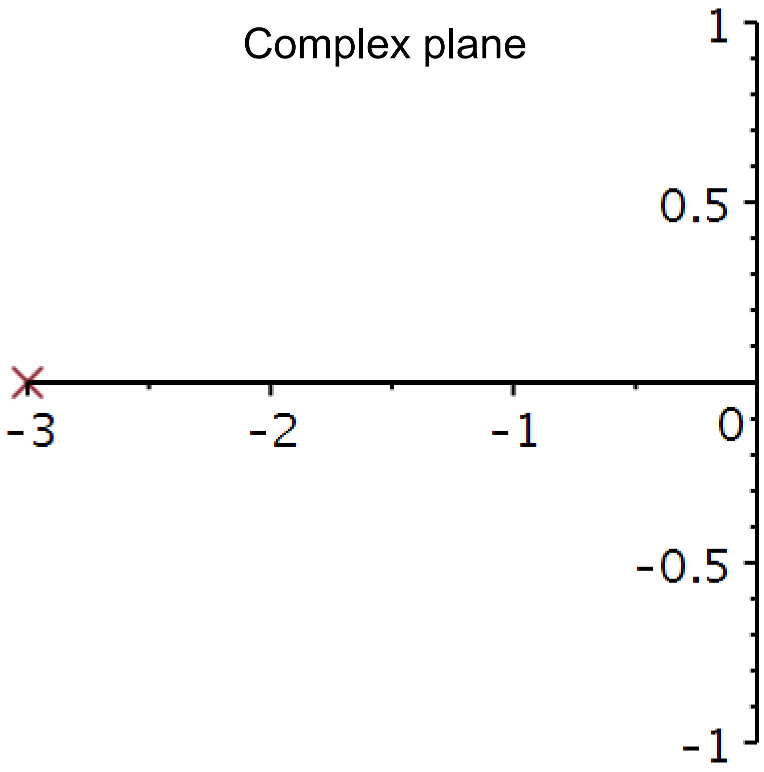
Two real poles

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = x(t)$$



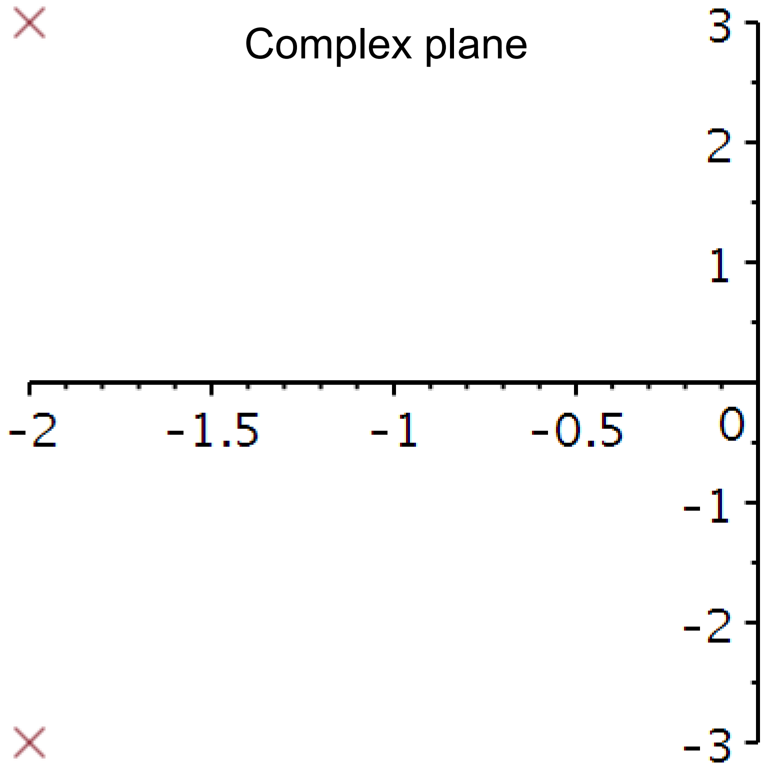
A double pole

$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = x(t)$$

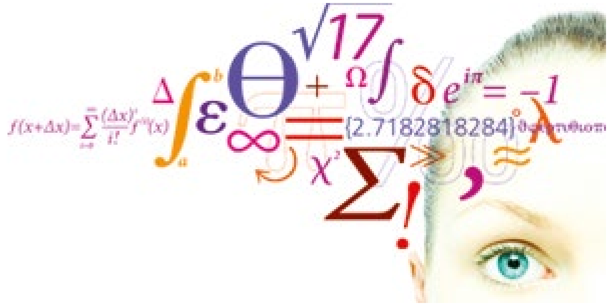


Complex conjugated poles

$$\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = x(t)$$



Break 10 min.

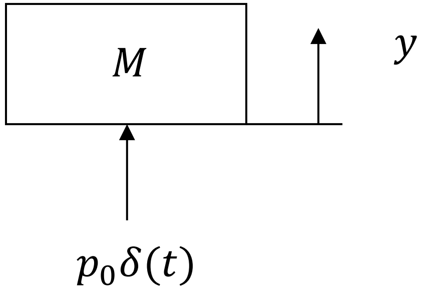


Impulse response

Video 3

Effect of an impulse disturbance

Imagine that we hit the mass M with a hammer from beneath. The hammer will act with an impulse p_0 on the mass and the momentum of the mass will increase by an amount p_0:



$$p(t) = p_0 u(t) = Mv(t) = M\dot{y}(t)$$

$$\frac{dp}{dt} = M\dot{v} = M\ddot{y} = p_0\dot{u}(t) = p_0\delta(t)$$

$$F = M\ddot{y} = p_0\delta(t) \;\; \wedge \;\; \dot{y}(0_-) = 0 \;\; \wedge \;\; y(0_-) = 0$$

$$M\dot{y}(0_+) - M\dot{y}(0_-) = \int_{0_-}^{0_+} M\ddot{y}dt = p_0 \underbrace{\int_{0_-}^{0_+} \delta(t)dt}_{=1} = p_0$$

$$\underbrace{\dot{y}(0_+) - \dot{y}(0_-)}_{velocity\ change} = \frac{p_0}{M}$$

$$\int_{0_-}^{0_+} \dot{y}dt = \underbrace{y(0_+) - y(0_-)}_{position\ change} = \frac{p_0}{M}(0_+ - 0_-) = 0$$

Observation:

The effect of an impulse disturbance is an instantaneous change in velocity $\dot{y}$.

Because the change in velocity is finite, there is not enough time for a change in position y.

Response to sudden change on input

The differential equation and its initial conditions shown here describes a second order system, which is at rest for $t < 0$.

$$\ddot{y} + a_1\dot{y} + a_0 y = x(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

If the input $x(t)$ changes instantaneously, the highest derivative must be able to follow this instantaneous change.

Fx, we cannot have a change in position without a velocity, and we cannot have a change in velocity without an acceleration.

An instantaneous change in velocity requires an infinitely large acceleration.

An instantaneous change in position requires an infinitely large velocity.

Response to sudden change on input

Here the input is an impulse function.

At t = 0, acceleration becomes an impulse, velocity a step function and position becomes a ramp function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \delta(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\, t = 0: \qquad \ddot{y} \propto \delta(t) \qquad \dot{y} \propto u(t) \qquad y \propto r(t)$$

Here the input is the derivative of the impulse function.

At t = 0, acceleration becomes the derivative of the impulse, velocity an impulse function and position becomes a step function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \dot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\, t = 0: \qquad \ddot{y} \propto \dot{\delta}(t) \qquad \dot{y} \propto \delta(t) \qquad y \propto u(t)$$

Here the input is the 2nd derivative of the impulse function.

At t = 0, acceleration becomes the 2nd derivative of the impulse, velocity is the 1st derivative of the impulse function and position becomes an impulse function.

$$\ddot{y} + a_1\dot{y} + a_0 y = \ddot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$@\, t = 0: \qquad \ddot{y} \propto \ddot{\delta}(t) \qquad \dot{y} \propto \dot{\delta}(t) \qquad y \propto \delta(t)$$

Effect at t = 0 of an impulse disturbance

$$\ddot{y} + a_1\dot{y} + a_0 y = \delta(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

At t = 0, acceleration behaves like an impulse:

$$\ddot{y}(t) = \delta(t)$$

At $t = 0$:

The acceleration is an impulse, the velocity changes as a step function, and the position changes as a ramp function.

$$\int_{0_-}^{t} \ddot{y}(s)ds = \quad \dot{y}(t) - \dot{y}(0_-) = \int_{0_-}^{t} \delta(s)ds = u(t)$$

$$\int_{0_-}^{t} \dot{y}(s)ds = \quad y(t) - y(0_-) = \int_{0_-}^{t} u(s)\, ds = r(t)$$

We observe that the effect of a unit impulse is an instantaneous unit jump in velocity:

$$\dot{y}(0_+) - \dot{y}(0_-) = 1$$

$$y(0_+) - y(0_-) = r(0_+) = 0$$

Effect at t = 0 of an impulse disturbance

$$\ddot{y} + a_1\dot{y} + a_0 y = \dot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

At $t = 0$:

Acceleration is the derivative of the impulse function, velocity $\dot{y}$ is an impulse function, and position is a step function.

$$\int_{0_-}^{t} \ddot{y}(s)ds = \dot{y}(t) - \dot{y}(0_-) = \int_{0_-}^{t} \dot{\delta}(s)ds$$

$$\dot{y}(t) - \dot{y}(0_-) = \delta(t)$$

$$\int_{0_-}^{t} \dot{y}(s)ds = y(t) - y(0_-) = \int_{0_-}^{t} \delta(s)ds = u(t)$$

We observe that the effect of a unit impulse change in velocity creates an instantaneous unit jump in position:

$$y(0_+) - y(0_-) = 1$$

Effect at t = 0 of an impulse disturbance

$$\ddot{y} + a_1\dot{y} + a_0 y = \ddot{\delta}(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

At $t = 0$:

acceleration is the 2nd derivative of the impulse function, velocity $\dot{y}$ is the 1st derivative of the impulse function, and position is an impulse function.

$$\int_{0_-}^{t} \ddot{y}(s)ds = \dot{y}(t) - \dot{y}(0_-) = \int_{0_-}^{t} \ddot{\delta}(s)ds$$

$$\dot{y}(t) - \dot{y}(0_-) = \dot{\delta}(t)$$

$$\int_{0_-}^{t} \dot{y}(s)ds = y(t) - y(0_-) = \int_{0_-}^{t} \dot{\delta}(s)ds = \delta(t)$$

We observe that if a system gets a 2nd derivative of the impulse as input, an impulse will occur on the output.

$$y(t) - y(0_-) = \delta(t)$$

Initial conditions for response to normalized impulse perturbation

An impulse perturbation produces an instant change in system state.

$$\ddot{y} + a_1\dot{y} + a_0 y = \delta(t), \qquad \dot{y}(0_-) = 0, y(0_-) = 0$$

$$\Rightarrow \dot{y}(0_+) - \dot{y}(0_-) = 1$$
$$\Rightarrow y(0_+) - y(0_-) = 0$$

Thus, for a system at rest, we can represent the impulse perturbation by specifying the initial conditions:

$$\ddot{y} + a_1\dot{y} + a_0 y = 0 \qquad \dot{y}(0_+) = 1, y(0_+) = 0$$

For an n'th order system:

$$y^{(n-1)}(0_+) = 1 \quad y^{(n-2)}(0_+) = y^{(n-3)}(0_+) = \cdots = y(0_+) = 0$$

Observation:

- If we have a differential equation where the forcing function $x(t) = \delta(t)$, then we have in principle an inhomogeneous differential equation.
- We have seen that for a 2nd order system the effect of a unit impulse is a sudden increase in $\dot{y}(t)$ by 1. If $\dot{y}(0_-) = 0$, then $\dot{y}(0_+) = 1$.
- Instead of solving an inhomogeneous differential equation, we will solve the corresponding homogeneous equation, using the initial conditions caused by the unit impulse.

Solving for impulse response

When seeking the response to an impulse perturbation, we replace the input function by a unit impulse function.

We can solve this equation by assigning appropriate initial conditions to the homogeneous equation. Subscript n refers to the *natural* response.

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = x(t)$$

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = \delta(t)$$

$$\ddot{y}_n(t) + a_1\dot{y}_n(t) + a_0 y_n(t) = 0 \qquad y(0_+) = 0, \dot{y}(0_+) = 1$$

The impulse response is the solution to a homogeneous differential equation:

It plays a vital role and is by convention denoted h.

$$h(t) \stackrel{\text{def}}{=} y_n(t)u(t) = \sum_{j=1}^{n} C_j e^{\lambda_j t}\,; t \geq 0_+, m < n$$

We have considered only the case when $b_0 = 1$. We need to generalize the result to a general operator polynomial $P(D)x(t)$

Solving for impulse response

Unscaled input:

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = x(t) \qquad y(0_+) = 0, \dot{y}(0_+) = 1$$

$$h(t) = y_n(t)u(t)$$

If we scale the input by a factor b_0, we must scale the output by that same factor b_0:

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = b_0 x(t)$$

$$h(t) = b_0 y_n(t)u(t)$$

Solving for impulse response

Unscaled input:

$$\ddot{y}_1(t) + a_1\dot{y}_1(t) + a_0 y_1(t) = x(t) \qquad h_1(t) = y_1(t)u(t)$$

If we differentiate on the rhs, we must differentiate on the lhs:

$$\dddot{y}_1(t) + a_1\ddot{y}_1(t) + a_0\dot{y}_1(t) = \dot{x}(t)$$

$$\ddot{y}_2(t) + a_1\dot{y}_2(t) + a_0 y_2(t) = \dot{x}(t) \qquad h_2(t) = y_2(t)u(t)$$

$$\ddot{y}_2(t) + a_1\dot{y}_2(t) + a_0 y_2(t) = b_1\dot{x}(t) \qquad h_2(t) = b_1 y_2(t)u(t)$$

$$y_2(t) = \dot{y}_1(t) \qquad h_2(t) = b_1\frac{d}{dt}\big(y_1(t)u(t)\big)$$

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0 y(t) = b_1\dot{x}(t) + b_0 x(t) = P(D)x(t)$$

If the input is a linear combination, then the output will be a linear combination:

$$h(t) = b_1\frac{d}{dt}\big(y_n(t)u(t)\big) + b_0 y_n(t)u(t) = P(D)\big(y_n(t)u(t)\big)$$

Solving for impulse response

If we have a differential equation with a derivative on the right-hand side, we will never attempt to solve this.

$$\ddot{y}_2(t) + a_1\dot{y}_2(t) + a_0 y_2(t) = b_1\dot{x}(t)$$

$$\ddot{y}_2(t) + a_1\dot{y}_2(t) + a_0 y_2(t) = b_1\dot{\delta}(t)$$

We will solve for the simpler case, and post-manipulate the solution to fit the equation at hand:

$$\ddot{y}_1(t) + a_1\dot{y}_1(t) + a_0 y_1(t) = \delta(t)$$

$$\ddot{y}_1(t) + a_1\dot{y}_1(t) + a_0 y_1(t) = 0 \qquad y_1(0_+) = 0, \dot{y}_1(0_+) = 1$$

$$h_2(t) = b_1 \frac{d}{dt}\left(y_1(t)u(t)\right)$$

Solving for impulse response

We can split the differential operation in two terms:

$$h(t) = P(D)\big(y_n(t)u(t)\big) = (b_m D^m + \cdots + b_1 D + b_0)\big(y_n(t)u(t)\big)$$
$$= b_m D^m\big(y_n(t)u(t)\big) + b_{m-1}D^{m-1}\big(y_n(t)u(t)\big) \cdots + b_0\big(y_n(t)u(t)\big)$$

$$(b_m D^m + \cdots + b_1 D + b_0)\big(y_n(t)u(t)\big) = b_0 y_n(t)u(t) + b_m y_n(t)D^m\big(u(t)\big) + b_m u(t)D^m\big(y_n(t)\big) +$$
$$\cdots + b_1 y_n(t)D\big(u(t)\big) + b_1 u(t)D\big(y_n(t)\big)$$

The blue terms are zero.

$$t > 0: u(t) = 1, \qquad y_n(t)D^m\big(u(t)\big) = 0$$

$$t = 0: y_n(0) = 0, \qquad y_n(0)D^m\big(u(t)\big) = 0$$

$$h(t) = P(D)\big(y_n(t)u(t)\big) = u(t)P(D)\big(y_n(t)\big)$$

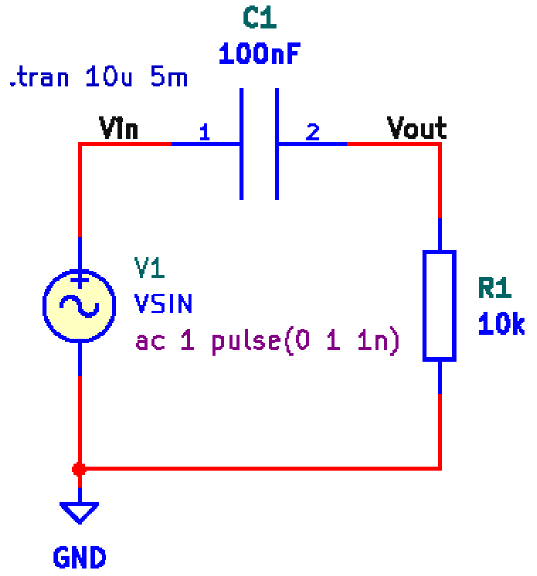
Solving for impulse response

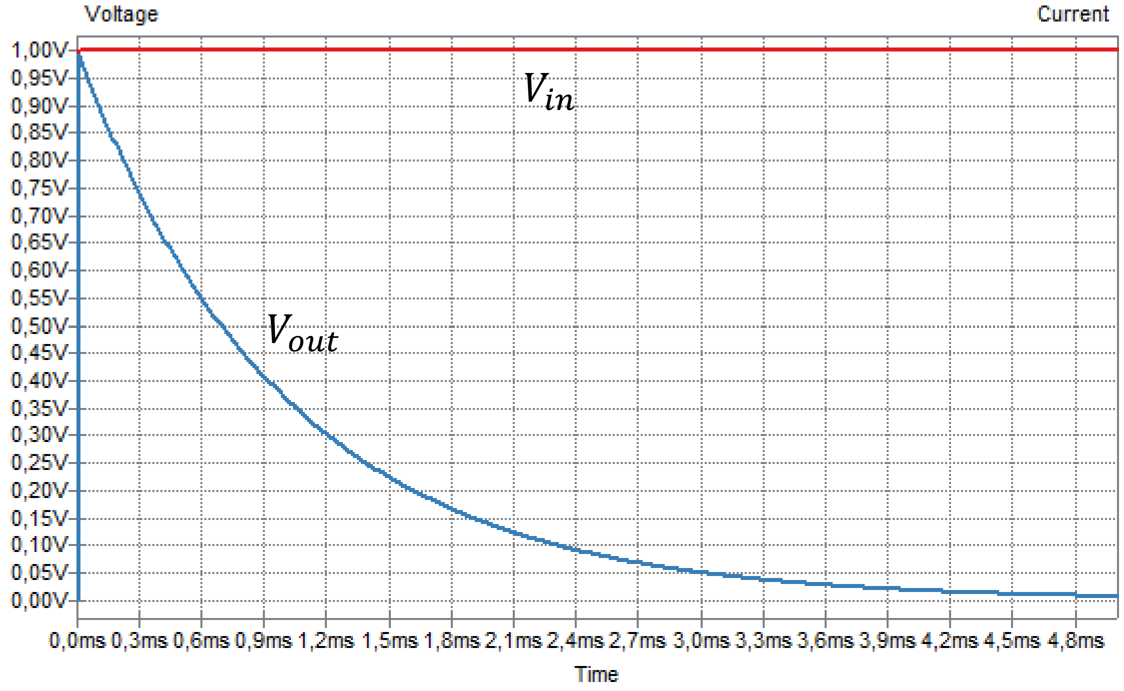
The present circuit will respond with a sudden jump on the output if a sudden jump is applied at the input.

We recall that the voltage drop across a capacitor cannot change instantaneously. The unit step response seen on the output of this RC circuit clearly shows that the output response includes a unit step from zero to one at $t = 0$.

Because the unit impulse function is the derivative of the unit step function, the unit impulse response will also be the derivative of the unit step response.

If we are to differentiate the blue curve at $t = 0$, we need a unit impulse function.





Solving for impulse response

There are systems, like the circuit on the previous slide, where a sudden change of input results in a synchronous sudden change on the output.

$m = n$

$$(D^n + a_{n-1}D^{n-1} + \ldots + a_1 D + a_0)h(t) = (b_n D^n + b_{n-1}D^{n-1} + \ldots + b_1 D + b_0)\delta(t)$$

In the differential equation shown here, the right-hand side contains the n'th derivative of the impulse function.

This requires the impulse response to also include an impulse function, such that the n'th derivative of the impulse function is present on both sides.

$$h(t) = A_0\delta(t) + \big(P(D)y_n(t)\big)u(t)$$

Matching the terms with the highest order of derivative of $\delta(t)$ we obtain:

$$D^n\big(A_0\delta(t)\big) = b_n D^n \delta(t) \Rightarrow A_0 = b_n$$

When the order m of the $P(D)$ operator polynomial equals the order n of the $Q(D)$ operator polynomial, an impulse function must be included in the impulse response.

$$h(t) = b_n\delta(t) + \big(P(D)y_n(t)\big)u(t)$$

Examples

Let us find the natural response:

$$\ddot{y}_n + 4\dot{y}_n + 4y_n = 0, \qquad y_n(0_+) = 0, \dot{y}_n(0_+) = 1$$

We solve the homogeneous equation with special initial conditions.

$$\lambda^2 + 4\lambda + 4 = (\lambda + 2)^2 = 0 \Rightarrow \lambda = -2$$

$$y_n(t) = (C_1 + C_2 t)e^{-2t}$$

What type of roots will we see?

A. Two real roots
B. A double root
C. Complex conjugated roots

$$\dot{y}_n(t) = \big(-2C_1 + C_2(1 - 2t)\big)e^{-2t}$$

$$y_n(0_+) = 0 \Rightarrow C_1 = 0 \wedge \dot{y}(0_+) = 1 \Rightarrow C_2 = 1$$

$$\frac{a_1^2}{a_0} = \frac{4^2}{4} = 4$$

$$y_n(t) = te^{-2t}; t \geq 0$$

$$h(t) = b_n\delta(t) + \big(P(D)y_n(t)\big)u(t), m \leq n$$

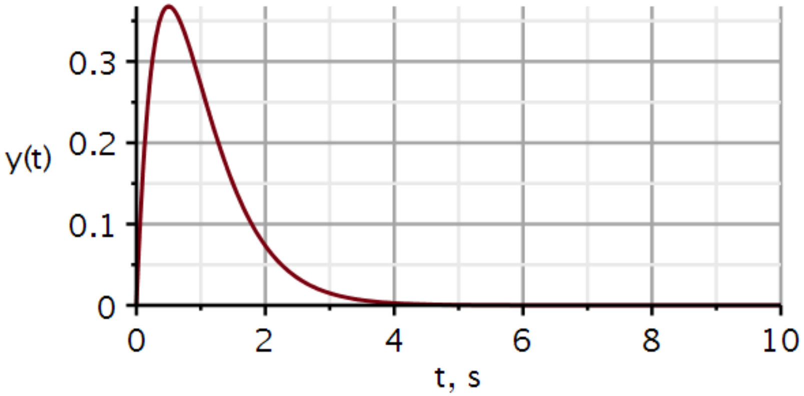
Without knowing the $P(D)$ polynomial, we cannot write the final form of the impulse response.

Examples

$$\ddot{y} + 4\dot{y} + 4y = 2x$$

$$m < n$$

$$h(t) = [P(D)te^{-2t}]u(t)$$
$$= [2te^{-2t}]u(t)$$

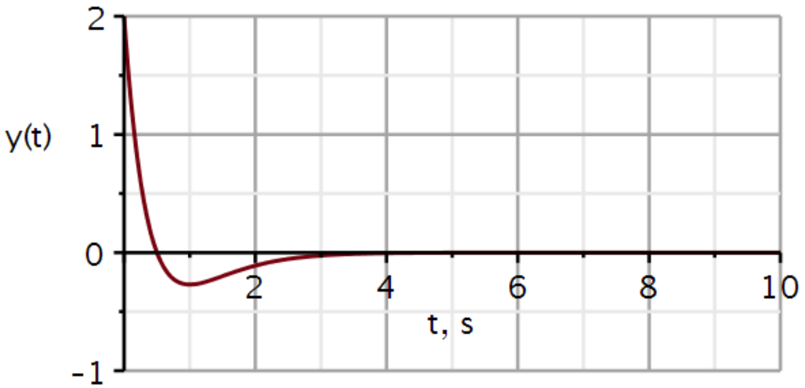


$$@t = 0: y(t) \propto r(t)$$

$$\ddot{y} + 4\dot{y} + 4y = 2\dot{x}$$

$$m < n$$

$$h(t) = [P(D)te^{-2t}]u(t)$$
$$= \left[2\frac{d}{dt}(te^{-2t})\right]u(t)$$
$$= 2[1 - 2t]e^{-2t}u(t)$$

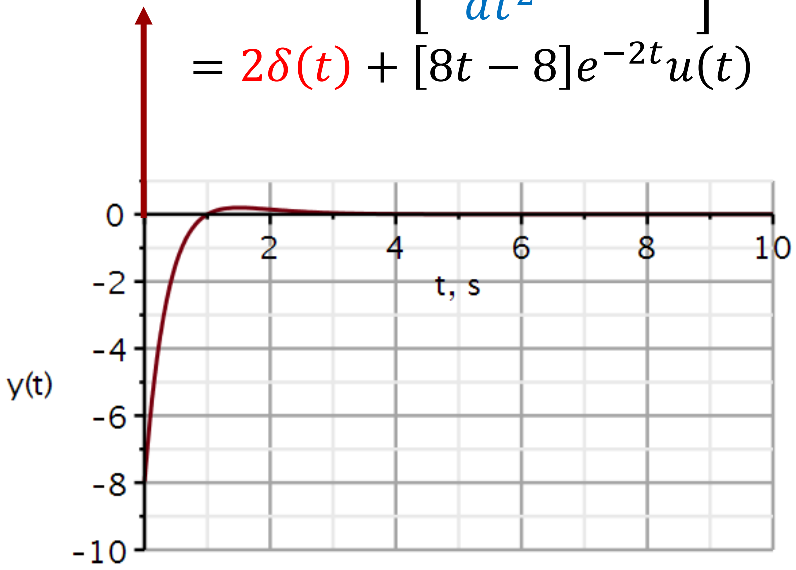


$$@t = 0: y(t) \propto u(t)$$

$$\ddot{y} + 4\dot{y} + 4y = 2\ddot{x}$$

$$m = n$$

$$h(t) = 2\delta(t) + [P(D)te^{-2t}]u(t)$$
$$= 2\delta(t) + \left[2\frac{d^2}{dt^2}(te^{-2t})\right]u(t)$$
$$= 2\delta(t) + [8t - 8]e^{-2t}u(t)$$



$$@t = 0: y(t) \propto \delta(t)$$

Impulse response – Two real roots

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = x(t)$$

$$\ddot{y}_n(t) + 3\dot{y}_n(t) + 2y_n(t) = 0, \dot{y}_n(0_+) = 1, y_n(0_+) = 0$$

$$Q(\lambda) = \lambda^2 + 3\lambda + 2 = 0$$

$$Q(\lambda) = (\lambda + 2)(\lambda + 1) = 0$$
$$\Leftrightarrow \lambda_1 = -2 \wedge \lambda_2 = -1$$

$$y_n(t) = C_1 e^{-2t} + C_2 e^{-t}; t \geq 0$$

$$C_2 = \frac{\lambda_1 y_n(0_+) - \dot{y}_n(0_+)}{\lambda_1 - \lambda_2} = \frac{-2 \cdot 0 - 1}{-2 - (-1)} = 1$$

$$C_1 = y_n(0_+) - C_2 = 0 - 1 = -1$$

$$y_n(t) = (e^{-t} - e^{-2t})$$

$$h(t) = (e^{-t} - e^{-2t})u(t)$$

Analytic impulse response

Initial conditions

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

Differential equation

$ode := diff(y(t), t, t) + 3 \cdot diff(y(t), t) + 2 \cdot y(t) = 0 :$

$solution := evalf(dsolve(\{ode, ics\}, y(t)))$

$solution := y(t) = -1.\, \mathrm{e}^{-2.\, t} + \mathrm{e}^{-1.\, t}$

$h1 := t \rightarrow (\mathrm{e}^{-1.\, t} - \mathrm{e}^{-2.\, t}) \cdot u(t) :$

Impulse response – One double root

$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = 0; y(0) = 0, \dot{y}(0) = 1$$

$$Q(\lambda) = \lambda^2 + 6\lambda + 9 = 0$$

$$Q(\lambda) = (\lambda + 3)(\lambda + 3) = 0 \Leftrightarrow \lambda = -3$$

$$y_n(t) = (C_1 + C_2 t)e^{-3t}; t \geq 0$$

$$y_n(0) = C_1 = 0$$

$$\dot{y}_n(t) = (C_1\lambda + C_2 + C_2\lambda t)e^{\lambda t}$$

$$\dot{y}_n(0) = C_1\lambda + C_2 \Rightarrow C_2 = \dot{y}_n(0) - C_1\lambda$$
$$C_2 = 1 - 0(-3) = 1$$

$$y_n(t) = te^{-3t}$$

$$h(t) = te^{-3t}u(t)$$

Analytic impulse repsonse

$u := t \rightarrow \mathrm{Heaviside}(t) :$

Initial conditions

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

$$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$$

Differential equation

$ode := diff(y(t), t, t) + 6 \cdot diff(y(t), t) + 9 \cdot y(t) = 0 :$

$solution := evalf(dsolve(\{ode, ics\}, y(t)))$

$$solution := y(t) = \mathrm{e}^{-3.\,t}\, t$$

$h2 := t \rightarrow t \cdot \mathrm{e}^{-3.\,t} u(t) :$

Impulse response – Complex conjugated roots

$$\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = 0; y(0_+) = 0, \dot{y}(0_+) = 1$$

$$Q(\lambda) = \lambda^2 + 4\lambda + 13 = 0$$

$$Q(\lambda) = (\lambda + 2 + j3)(\lambda + 2 - j3) = 0 \Leftrightarrow \lambda = -2 \pm j3$$

$$y_n(t) = C_1 e^{-2t}\cos(3t) + C_2 e^{-2t}\sin(3t)\,; t \geq 0$$

$$y_n(0) = C_1 = 0$$

$$\dot{y}_n(t) = C_1\sigma e^{\sigma t}\cos(\omega t) - C_1\omega\sin(\omega t)$$
$$+C_2\sigma e^{\sigma t}\sin(\omega t) + C_2\omega e^{\sigma t}\cos(\omega t)$$

$$\dot{y}_n(0_+) = C_1\sigma + C_2\omega = 1 \Rightarrow C_2 = \frac{\dot{y}_0(0_+) - C_1\sigma}{\omega}$$

$$\Rightarrow C_2 = \frac{1 - 0(-2)}{3} = \frac{1}{3}$$

$$y_n(t) = \frac{1}{3}e^{-2t}\sin(3t)$$

$$h(t) = \frac{1}{3}e^{-2t}\sin(3t)\,u(t)$$

Analytic impulse response

Initial conditions

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

Differential equation

$ode := diff(y(t), t, t) + 4 \cdot diff(y(t), t) + 13 \cdot y(t) = 0 :$

$solution := evalf(dsolve(\{ode, ics\}, y(t)))$

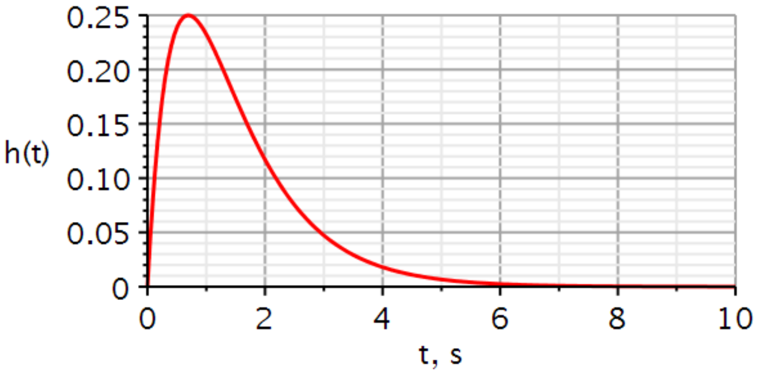
$solution := y(t) = 0.3333333333\,\mathrm{e}^{-2.\,t}\sin(3.\,t)$

$h3 := t \rightarrow 0.3333333333\,\mathrm{e}^{-2.\,t}\sin(3.\,t) \cdot u(t)$

Impulse responses

Two real poles

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = \delta(t)$$

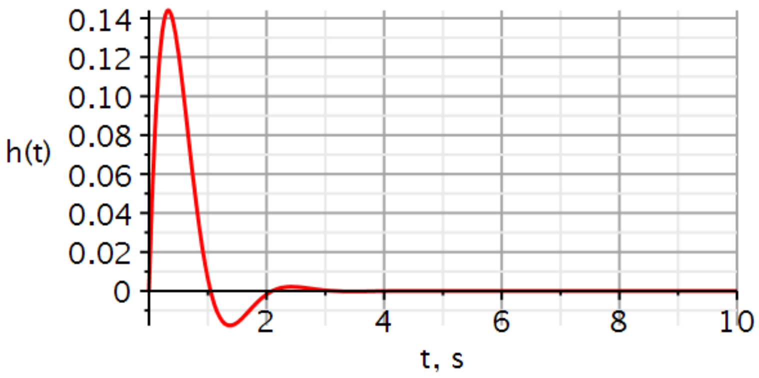


A double pole

$$\ddot{y}(t) + 6\dot{y}(t) + 9y(t) = \delta(t)$$

Complex conjugated poles

$$\ddot{y}(t) + 4\dot{y}(t) + 13y(t) = \delta(t)$$



with(DynamicSystems) → *ImpulseResponsePlot*(*sys1*, 10, *color* = [*red*], *thickness* = 3, *axesfont* = ["Helvetica", "ROMAN", 18], *axis*[2] = [*thickness* = 2.5], *axis*[1] = [*mode* = *linear*, *thickness* = 2.5], *labels* = ["t, s" , "h(t)"], *labelfont* = ["HELVETICA", 18], *gridlines*, *size* = [600, 300])

Relationship between impulse response and step response

We obtain the step response $y_u(t)$ by applying the step function $u(t)$ as input.

If we differentiate the input signal, we must differentiate the output. Since the unit impulse function is the derivative of the unit step function**, the impulse response is the derivative of the step response**.

Hence, if we already have the step response, we can obtain the impulse response simply by differentiating the step response.

This is mostly **important in experimental determi-nation** of the impulse response. It is almost impossible to create a perfect impulse. It is easier to create a good step function, (e.g., flipping a switch), measure the step response, fit that to an equation, and differentiate that to obtain the impulse response.

$$Q(D)y(t) = P(D)x(t)$$

$$Q(D)y_u(t) = P(D)u(t)$$ step

$$Q(D)h(t) = P(D)\delta(t)$$ impulse

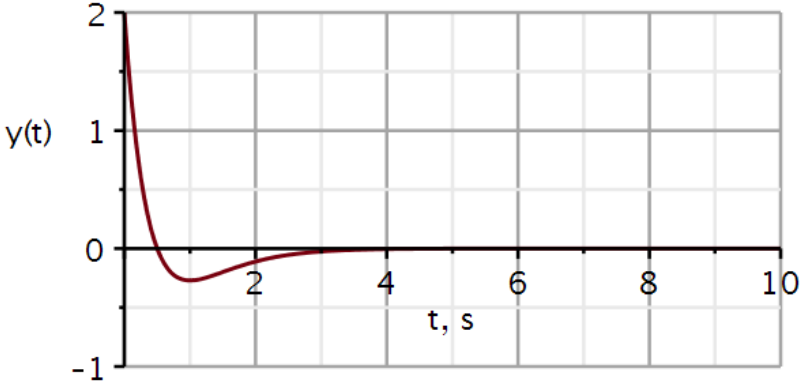
$$\delta(t) = \frac{du}{dt} \Rightarrow h(t) = \frac{dy_u(t)}{dt}$$

Quick quiz

Which impulse response belongs to which filter?

A. 2nd order Lowpass filter
B. 2nd order Highpass filter
C. 2nd order Bandpass filter

1



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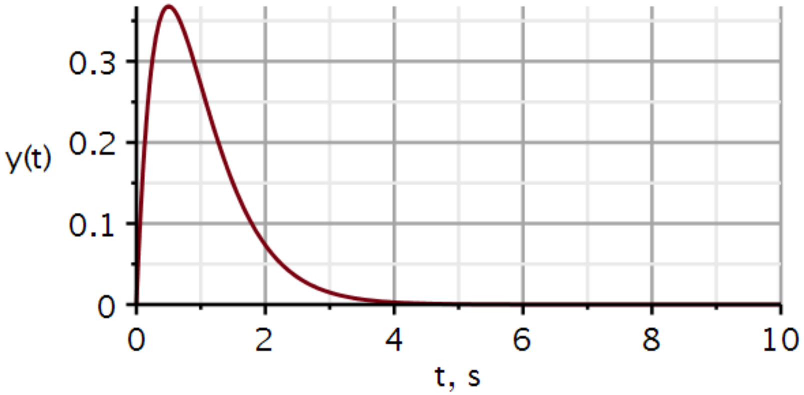

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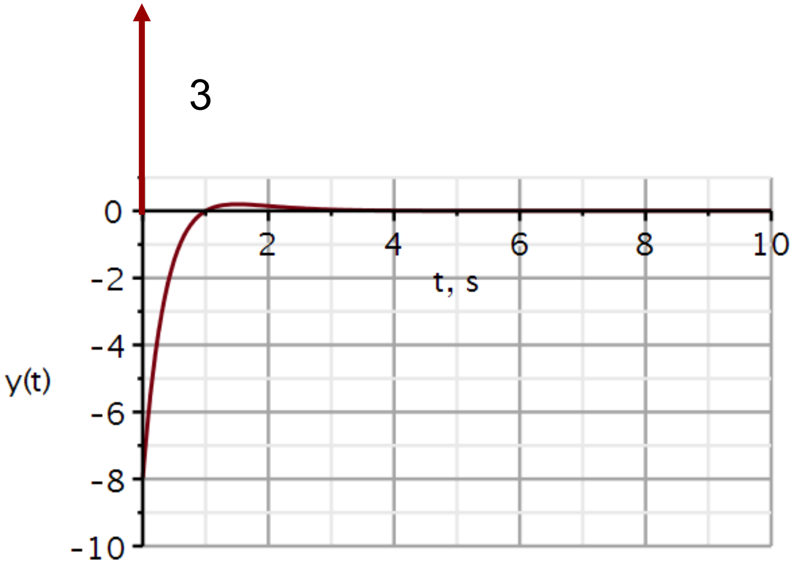
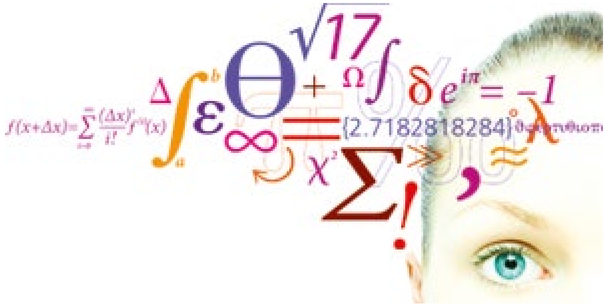
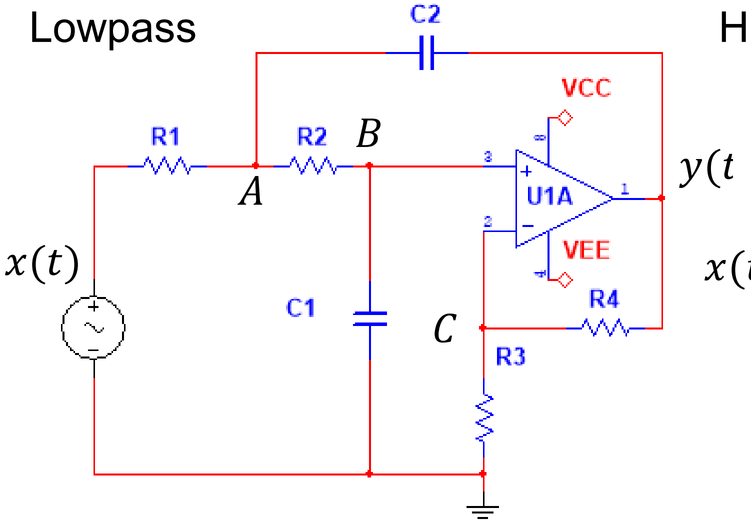




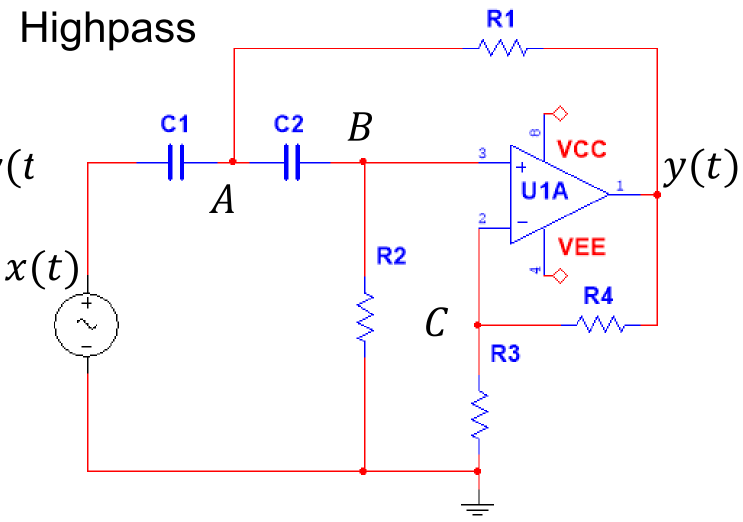
Problems

Sallen-Key filters used in this course

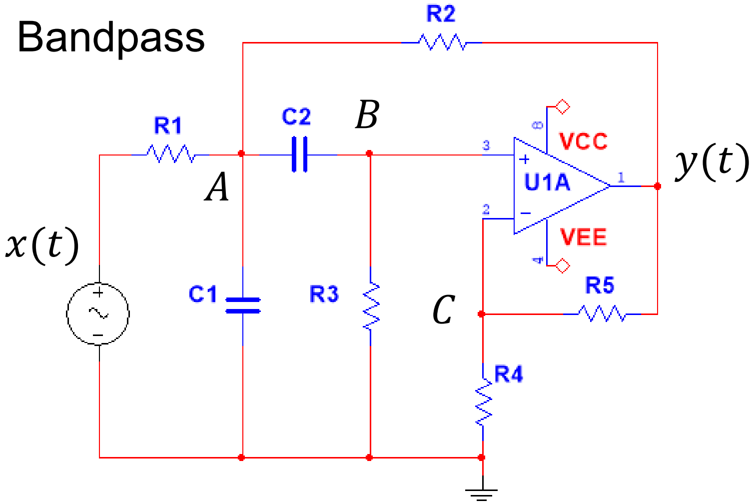
Lowpass



Highpass



Bandpass



Filter table

		Lowpass	Highpass	Bandpass
R_1	$k\Omega$	3.9894	1784.1	56.419
R_2	$k\Omega$	0.8865	892.06	34.117
R_3	$k\Omega$	1.0	1.0	149.72
R_4	$k\Omega$	1.0	1.0	1.0
R_5	$k\Omega$	–	–	1.0
C_1	nF	1795.2	1784.1	56.419
C_2	nF	398.94	3568.2	56.419
a_1		$2.83 \cdot 10^3$	$6.283 \cdot 10^{-1}$	$3.141 \cdot 10^1$
a_0		$3.95 \cdot 10^5$	$9.87 \cdot 10^{-2}$	$9.87 \cdot 10^4$
b_2		0	2	0
b_1		0	0	$6.283 \cdot 10^2$
b_0		$7.89 \cdot 10^5$	0	0

Filter 4: Lowpass filter – impulse response

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 1, show that the homogeneous differential equation has the solution: $y_0(t) = (2.116e^{-147.31\,t} - 0.116\,e^{-2680.2\,t})u(t)$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 311.8(e^{-147.31\,t} - e^{-2680.2\,t})u(t)$ and plot it.

Maple file available at DTU Learn.

Filter 5: Highpass filter – impulse response

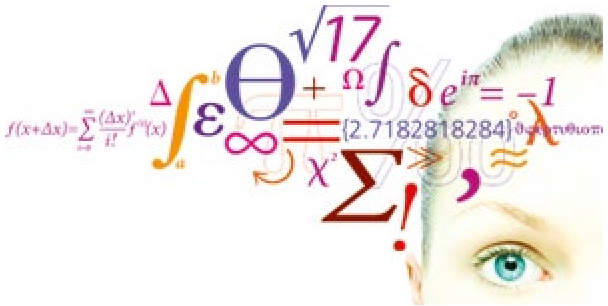
1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 2, show that the homogeneous differential equation has the solution: $y_0(t) = 0$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 2\delta(t) - 0.6283(2 - 031415\,t\,)e^{-031415\,t}u(t)$ and plot it.

Maple file NOT available at DTU Learn. Adapt from Filter 4 problem.

Filter 6: Bandpass filter – impulse response

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 3, show that the homogeneous differential equation has the solution: $y_0(t) = 0$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 628.3\ \mathrm{e}^{-15.7\,t}\cos(314\,t)u(t) - 31.4e^{-15.7\,t}\sin(314\,t)\,u(t)$ or equivalently $h(t) = 629.1\ e^{-15.7\,t}\cos(314\,t + 0.05)\ u(t)$ and plot it.

Maple file NOT available at DTU Learn. Adapt from Filter 4 problem.



Solutions

Filter 4: Lowpass filter – impulse response

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 1, show that the homogeneous differential equation has the solution: $y_0(t) = (2.116e^{-147.33\,t} - 0.116\,e^{-2680.2\,t})u(t)$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 311.8(e^{-147.31\,t} - e^{-2680.2\,t})u(t)$ and plot it.

Filter 4: Lowpass filter – impulse response (sol)

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.

   We know from problem Filter 1 that the system has a linear differential equation with constant coefficients. It is therefore linear and time-invariant.

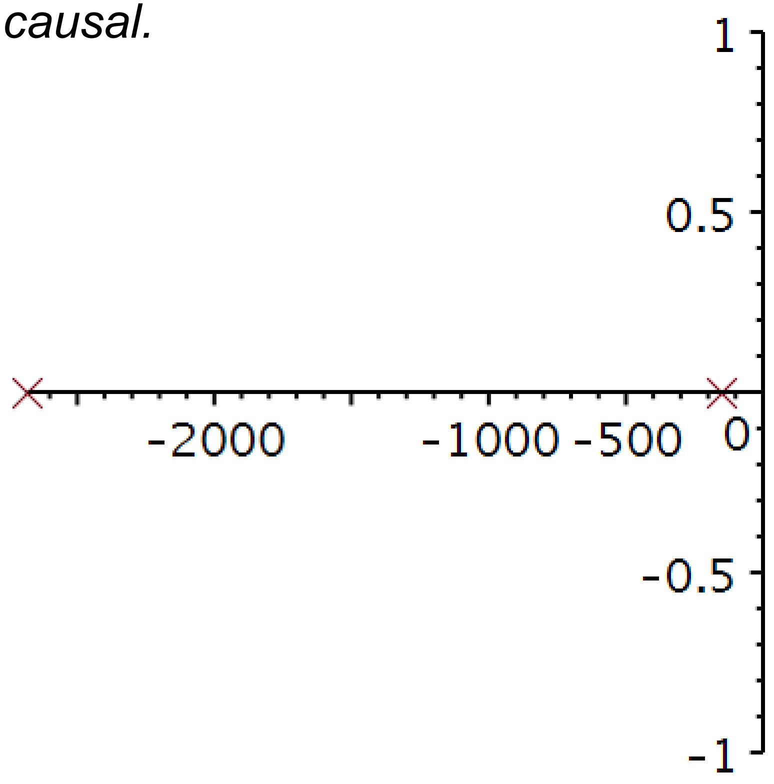
   The system does not require future input values and is therefore causal.

2. Draw the roots of the characteristic equation in the complex plane.

   Roots of characteristic equation

   $$solve\left(\lambda^2 + a1\cdot\lambda + a0 = 0, \lambda\right)$$

   $$-147.3061597, -2680.231742$$



3. Two real roots, hence an overdamped system.

4. Decomposition property? *The system is linear and time-invariant. Superposition holds and decomposition is possible.*

Filter 4: Lowpass filter – impulse response (sol)

5. Using the initial conditions derived in Filter Problem 1, show that the homogeneous differential equation has the solution: $y_0(t) = (2.116e^{-147.33\,t} - 0.116\,e^{-2680.2\,t})u(t)$

Initial conditions $y(0_-) = 2, \dot{y}(0_-) = 0$

$ics := y(0) = 2, \mathrm{D}(y)(0) = 0$

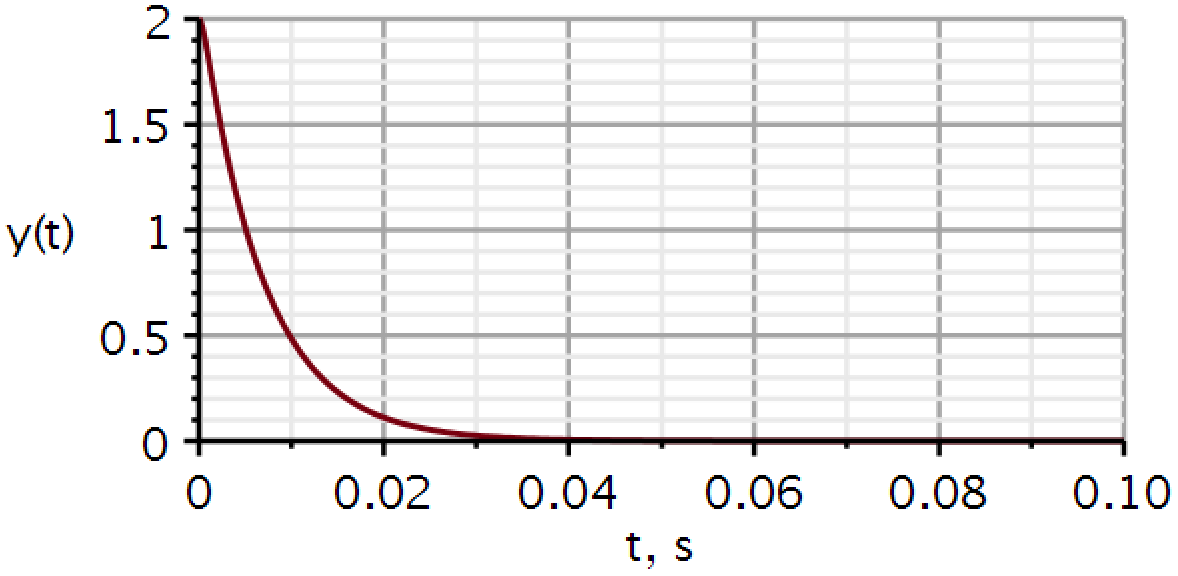
$$ics := y(0) = 2, \mathrm{D}(y)(0) = 0$$

Differential equation

$ode := diff(y(t), t, t) + a1 \cdot diff(y(t), t) + a0 \cdot y(t) = 0 :$
$solution := evalf(dsolve(\{ode, ics\}, y(t)), 5)$

$$solution := y(t) = 2.1163\,\mathrm{e}^{-147.33\,t} - 0.1163\,\mathrm{e}^{-2680.2\,t}$$

$y1 := t \rightarrow \left(2.1163\,\mathrm{e}^{-147.33\,t} - 0.1163\,\mathrm{e}^{-2680.2\,t}\right) \cdot \mathrm{Heaviside}(t)$



Filter 4: Lowpass filter – impulse response (sol)

6. Define the initial conditions appropriate for obtaining the impulse response.

$$y(0_+) = 0, \dot{y}(0_+) = 1$$

7. Show that the impulse response is: $h(t) = 311.8(e^{-147.31\,t} - e^{-2680.2\,t})u(t)$ and plot it.

Analytic impulse response

Initial conditions

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

$$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$$

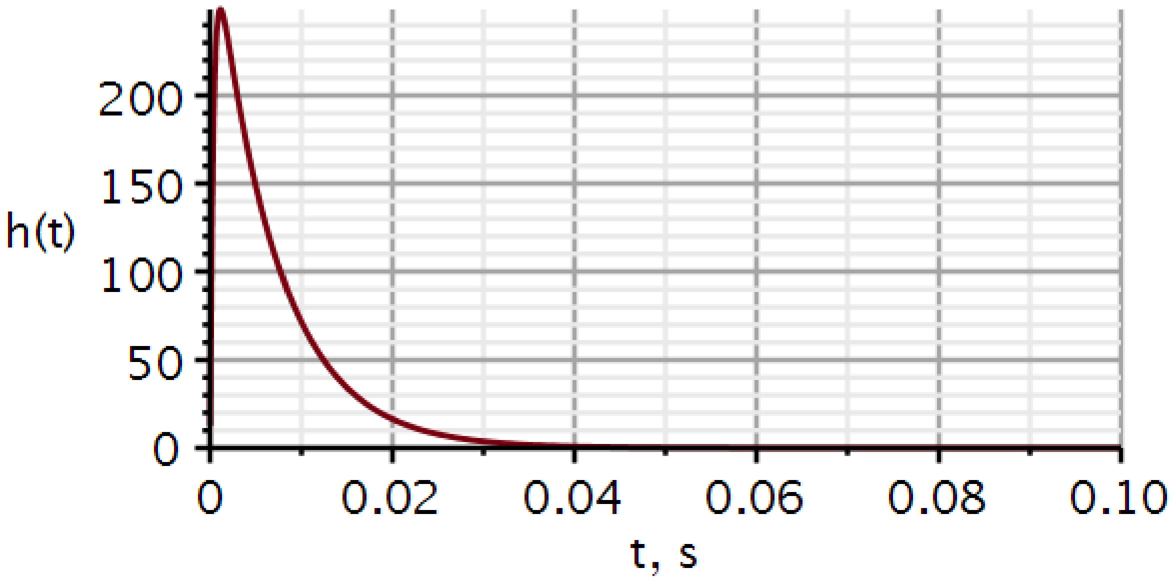
Differential equation

$ode := diff(y(t), t, t) + a1 \cdot diff(y(t), t) + a0 \cdot y(t) = 0:$
$solutionh := evalf(dsolve(\{ode, ics\}, y(t)))$

$$solutionh := y(t) = 0.0003948003868\, \mathrm{e}^{-147.3061599\,t} - 0.0003948003868\, \mathrm{e}^{-2680.231742\,t}$$

$assign(solutionh)$
$evalf(b0 \cdot y(t), 5) \cdot \mathrm{Heaviside}(t)$

$$\left(311.75\, \mathrm{e}^{-147.31\,t} - 311.75\, \mathrm{e}^{-2680.2\,t}\right) \mathrm{Heaviside}(t)$$



Filter 5: Highpass filter – impulse response

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 2, show that the homogeneous differential equation has the solution: $y_0(t) = 0$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 2\delta(t) - 0.6283(2 - 0.31415\, t\,)e^{-0.31415\, t}u(t)$ and plot it.

Filter 5: Highpass filter – impulse response (sol)

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.

   We know from problem Filter 2 that the system has a linear differential equation with constant coefficients. It is therefore linear and time-invariant.

   The system can be built and is therefore causal.

2. Draw the roots of the characteristic equation in the complex plane.

Roots of characteristic equation

$HProots := solve\left(\lambda^2 + a1\cdot\lambda + a0 = 0, \lambda\right)$

$HProots := -0.3141624760 + 0.001487536067\,\mathrm{I}, -0.3141624760 - 0.001487536067\,\mathrm{I}$

$p := -\mathrm{Re}(HProots[1])$

$p := 0.3141624760$

$a11 := 2\cdot p$

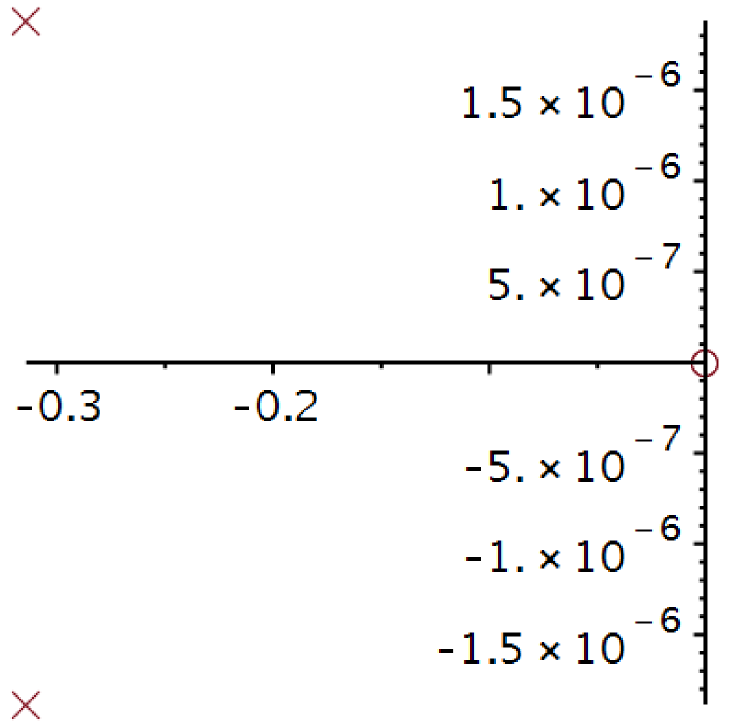
$a11 := 0.6283249520$

$a00 := p^2$

$a00 := 0.09869806133$

With decimal coefficients, it is impossible to get repeated poles, even if we ignore the imaginary part.

$p = \dfrac{\pi}{10}$



Filter 5: Highpass filter – impulse response (sol)

3. A double root, hence, a critically damped system.

4. The system is linear and time-invariant. Superposition holds and decomposition is possible.

5. Using the initial conditions derived in Filter Problem 2, show that the homogeneous differential equation has the solution: $y_0(t) = 0$

$$y(0_-) = 0, \dot{y}(0_-) = 0 \Rightarrow y_0(t) = 0$$

6. Define the initial conditions appropriate for obtaining the impulse response.

$$y(0_+) = 0, \dot{y}(0_+) = 1$$

Filter 5: Highpass filter – impulse response (sol)

7. Show that the impulse response is: $h(t) = 2\delta(t) - 0.6283(2 - 0.31415\,t)e^{-0.31415\,t}u(t)$ and plot it.

$$\ddot{y}(t) + 0.6248\dot{y}(t) + 0.09871y(t) = 0; y(0) = 0, \dot{y}(0) = 1$$

$$Q(\lambda) = \lambda^2 + 0.6248\lambda + 0.09871 = 0$$

$$Q(\lambda) \approx (\lambda + 0.3141624760)(\lambda + 0.3141624760) = 0 \Leftrightarrow \lambda = -\frac{\pi}{10}$$

$$y_n(t) = (C_1 + C_2 t)e^{-\lambda t}; t \geq 0$$

$$y_n(0_+) = C_1 = 0$$

$$\dot{y}_n(t) = (C_1\lambda + C_2 + C_2\lambda t)e^{\lambda t}$$

$$\dot{y}_n(0_+) = C_1\lambda + C_2 \Rightarrow C_2 = \dot{y}_n(0_+) - C_1\lambda$$
$$C_2 = 1 - 0(-3) = 1$$

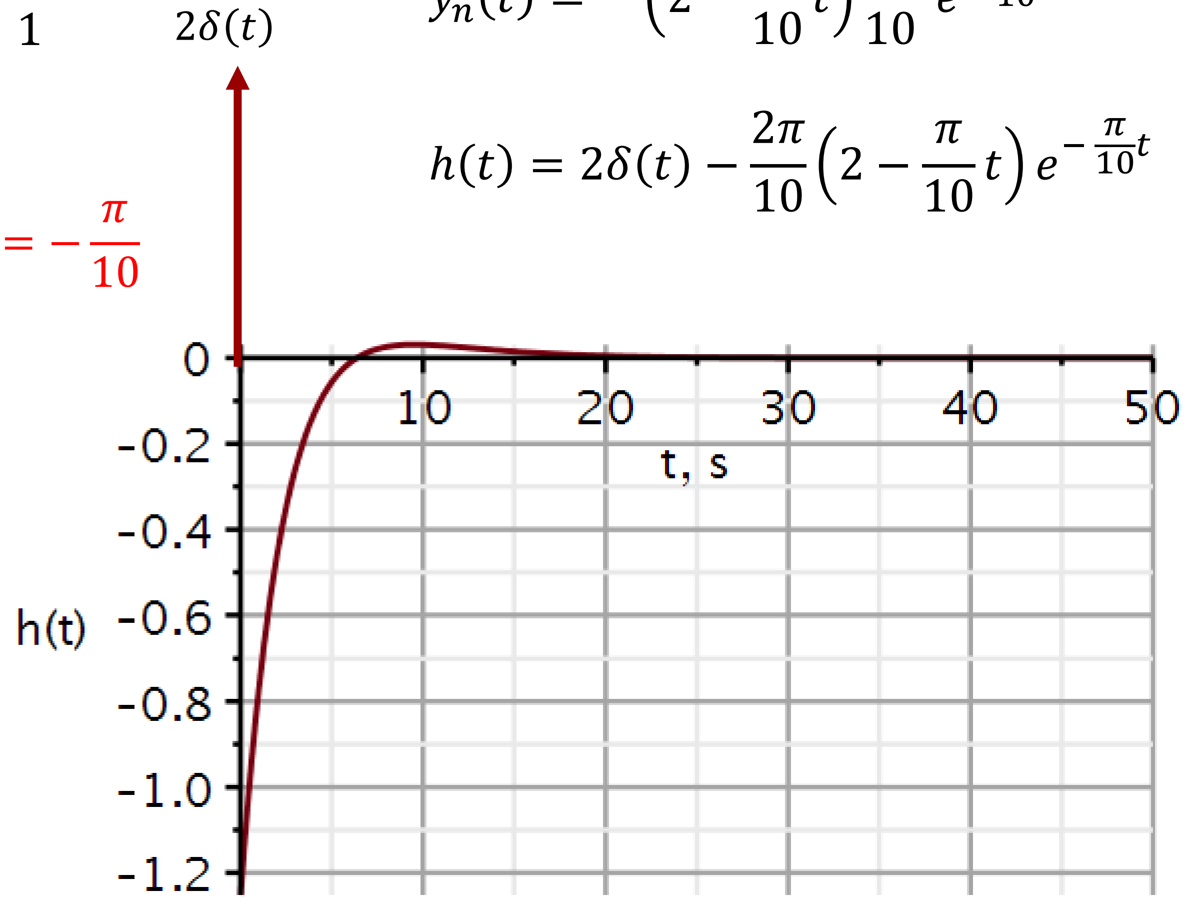
$$y_n(t) = te^{-\frac{\pi}{10}t}u(t)$$

$$h(t) = b_2\delta(t) + b_2\ddot{y}_n(t)$$

$$\dot{y}_{n(t)} = \left(1 - \frac{\pi}{10}t\right)e^{-\frac{\pi}{10}t}$$

$$\ddot{y}_n(t) = -\left(2 - \frac{\pi}{10}t\right)\frac{\pi}{10}e^{-\frac{\pi}{10}t}$$

$$h(t) = 2\delta(t) - \frac{2\pi}{10}\left(2 - \frac{\pi}{10}t\right)e^{-\frac{\pi}{10}t}$$



Filter 6: Bandpass filter – impulse response

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.
2. Draw the roots of the characteristic equation in the complex plane.
3. Based on the roots of the characteristic equation, classify the system as underdamped, critically damped, or overdamped. If the system is close to being critically damped (but not exactly), you can assume that is what was intended.
4. Check if the decomposition property holds for the system. If it does, the solution to the homogeneous differential equation is also the system's zero-input response
5. Using the initial conditions derived in Filter Problem 3, show that the homogeneous differential equation has the solution: $y_0(t) = 0$
6. Define the initial conditions appropriate for obtaining the impulse response.
7. Show that the impulse response is: $h(t) = 628.3\ \mathrm{e}^{-15.7\,t} \cos(314\ t) u(t) - 31.4 e^{-15.7\,t} \sin(314\ t)\, u(t)$ or equivalently $h(t) = 629.1\ e^{-15.7\,t} \cos(314\ t + 0.05)\ \ u(t)$ and plot it.

Filter 6: Bandpass filter – impulse response (sol)

1. Classify the system as linear/nonlinear, time-invariant/time-variant, causal/noncausal.

   We know from problem Filter 3 that the system has a linear differential equation with constant coefficients. It is therefore linear and time-invariant.

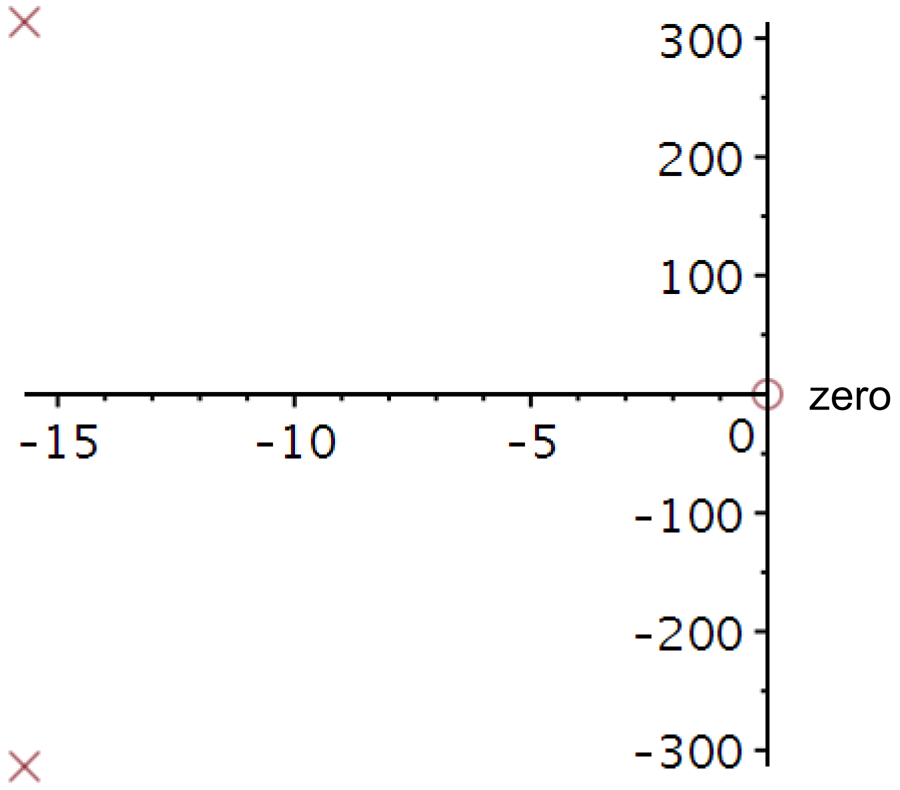
   The system can be built and is therefore causal.

2. Draw the roots of the characteristic equation in the complex plane.

Roots of characteristic equation

$solve(\lambda^2 + a1 \cdot \lambda + a0 = 0, \lambda)$

$-15.70298545 + 313.7646688\,I, \ -15.70298545 - 313.7646688\,I$



Filter 6: Bandpass filter – impulse response (sol)

3. Complex conjugated roots, hence, an underdamped system.
4. The system is linear and time-invariant. Superposition holds and decomposition is possible.
5. Using the initial conditions derived in Filter Problem 3, show that the homogeneous differential equation has the solution: $y_0(t) = 0$

$$y(0_-) = 0, \dot{y}(0_-) = 0 \Rightarrow y_0(t) = 0$$

6. Define the initial conditions appropriate for obtaining the impulse response.

$$y(0_+) = 0, \dot{y}(0_+) = 1$$

Filter 6: Bandpass filter – impulse response (sol)

7. Show that the impulse response is: $h(t) = 628.3\ \mathrm{e}^{-15.7\,t}\cos(314\,t)u(t) - 31.4e^{-15.7\,t}\sin(314\,t)\,u(t)$ or equivalently $h(t) = 629.1\,e^{-15.7\,t}\cos(314\,t + 0.05)\;u(t)$ and plot it.

$$C = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1}\left(-\frac{b}{a}\right)$$

Initial conditions

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

$ics := y(0) = 0, \mathrm{D}(y)(0) = 1$

Differential equation

$ode := diff(y(t), t, t) + a1 \cdot diff(y(t), t) + a0 \cdot y(t) = 0 :$

$solutionh := evalf(dsolve(\{ode, ics\}, y(t)))$

$solutionh := y(t) = 0.003187101990\ \mathrm{e}^{-15.70298545\,t}\sin(313.7646688\,t)$

$assign(solutionh)$

$evalf(b1 \cdot \dot{y}(t), 5) \cdot \mathrm{Heaviside}(t)$

$\left(-31.447\ \mathrm{e}^{-15.703\,t}\sin(313.76\,t) + 628.33\ \mathrm{e}^{-15.703\,t}\cos(313.76\,t)\right)\mathrm{Heaviside}(t)$

$A := 628.33 :$

$B := -31.447 :$

$\omega := 313.76 :$

$C := \sqrt{A^2 + B^2}$

$C := 629.1164461$

$\theta := \arctan(-B, A)$

$\theta := 0.05000681593$

$h3a := t \rightarrow C \cdot \mathrm{e}^{-15.703\,t} \cdot \cos(\omega \cdot t + \theta)$

$h3a := t \mapsto C \cdot \mathrm{e}^{(-1)\cdot 15.703 \cdot t} \cdot \cos(\omega \cdot t + \theta)$

$h3a(0.02)$

459.1482785

$h3b := t \rightarrow \left(-31.447\ \mathrm{e}^{-15.703\,t} \cdot \sin(313.76 \cdot t) + 628.33\ \mathrm{e}^{-15.703\,t} \cdot \cos(313.76 \cdot t)\right)$

$h3b(0.02)$

459.1482785

Filter 6: Bandpass filter – impulse response (sol)

```
ImpulseResponsePlot(sysBP, 0.1, color = [red], thickness = 3, axesfont
    = ["Helvetica", "ROMAN", 18], axis[2] = [thickness = 2.5], axis[1] = [mode
    = linear, thickness = 2.5], labels = ["t, s" , "h(t)"], labelfont = ["HELVETICA",
    18], gridlines, size = [600, 300])
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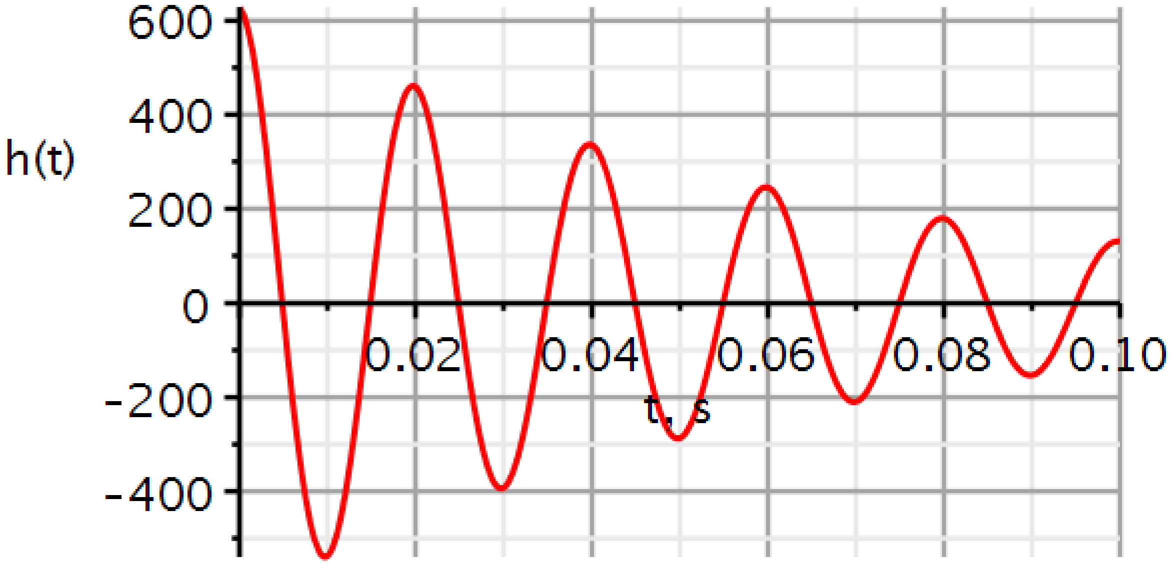

Analytical